

# BFS Traversal

Step 1: Start at A

Legend

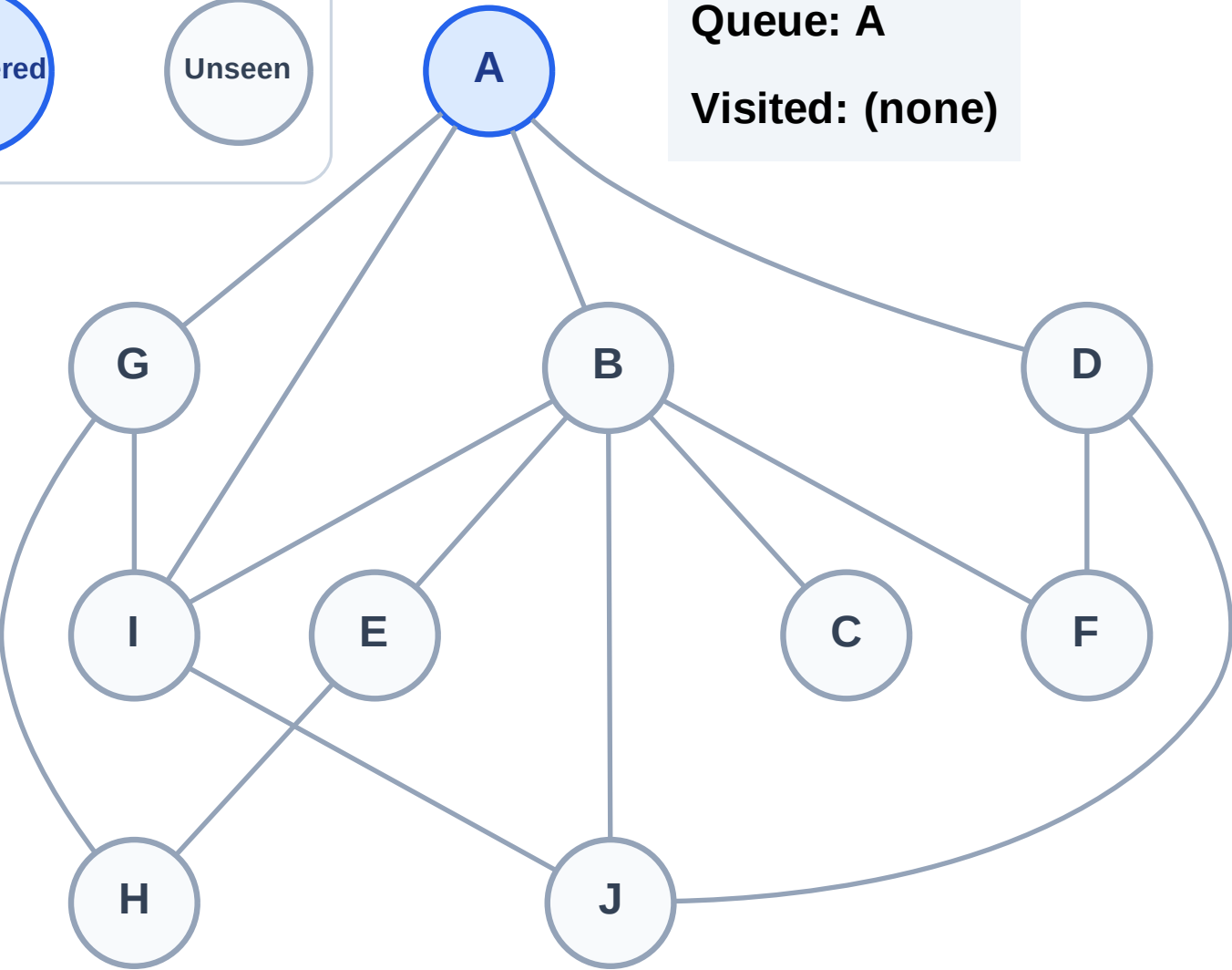
Complete

Processing

Discovered

Unseen

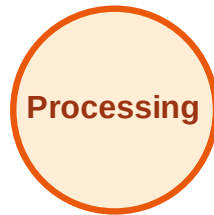
Queue: A  
Visited: (none)



# BFS Traversal

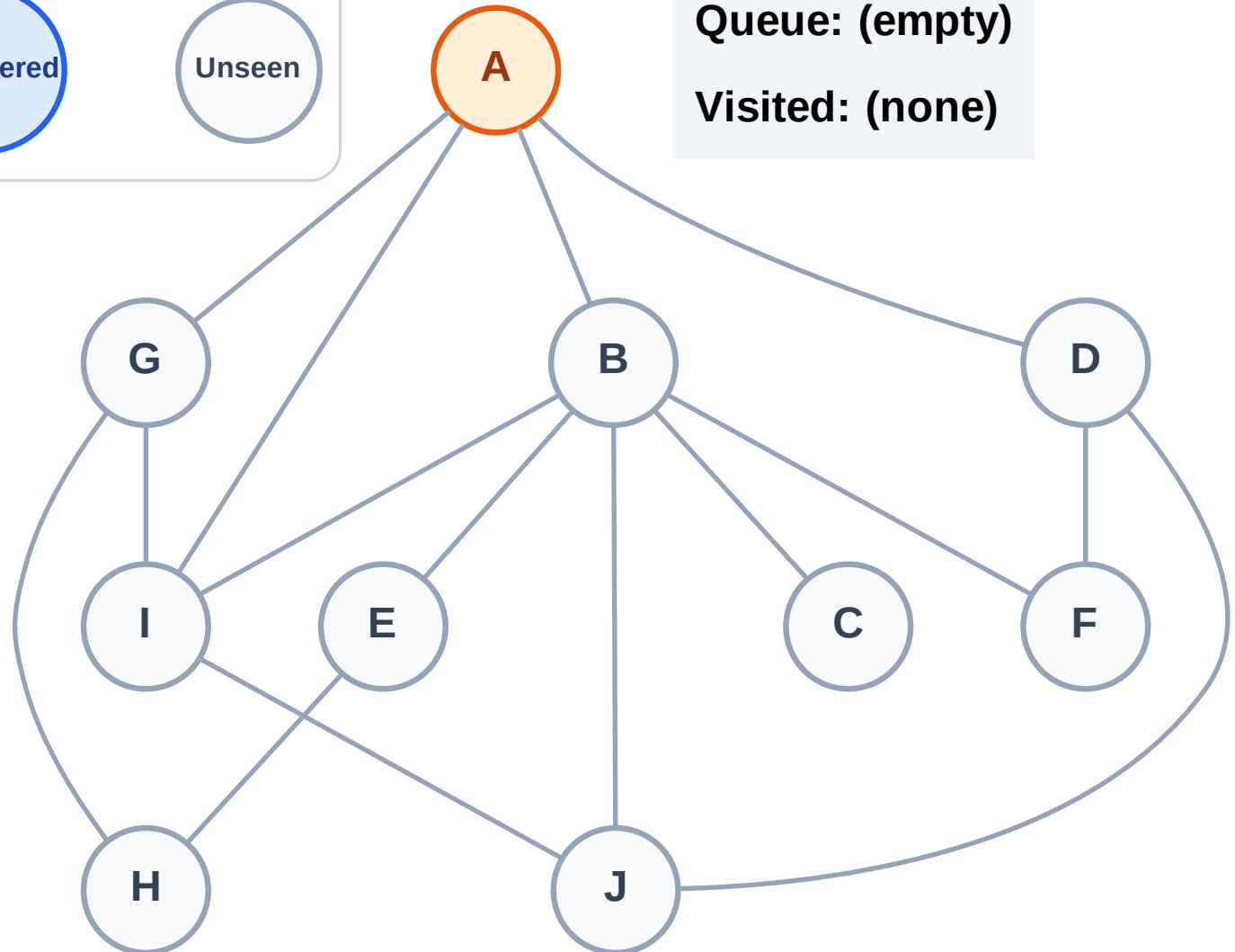
Step 2: Process node A

## Legend



Queue: (empty)

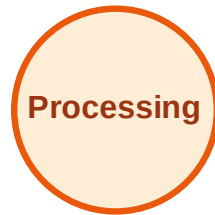
Visited: (none)



# BFS Traversal

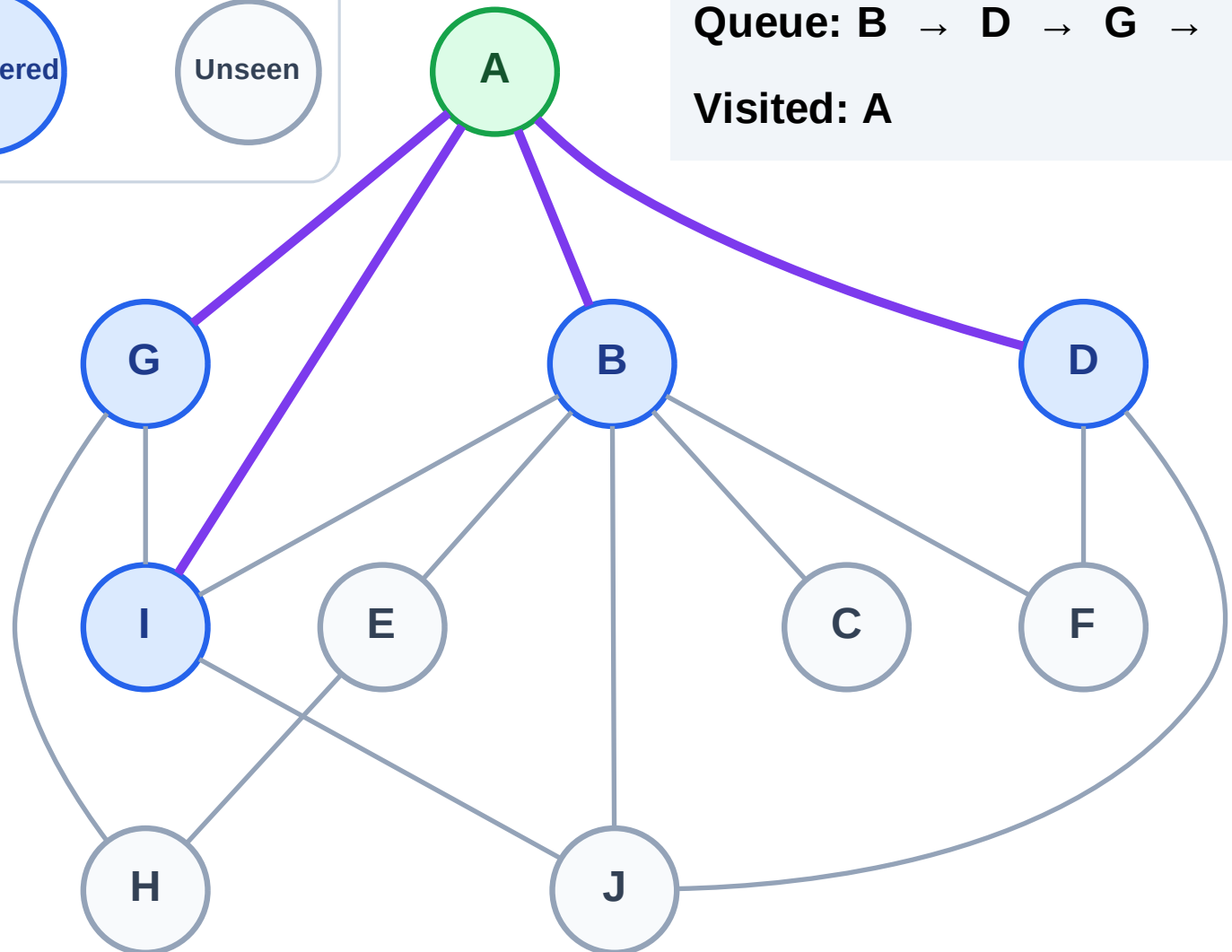
Step 3: Discover B, D, G, I from A

## Legend



Queue: B → D → G → I

Visited: A



# BFS Traversal

Step 4: Process node B

Legend

Complete

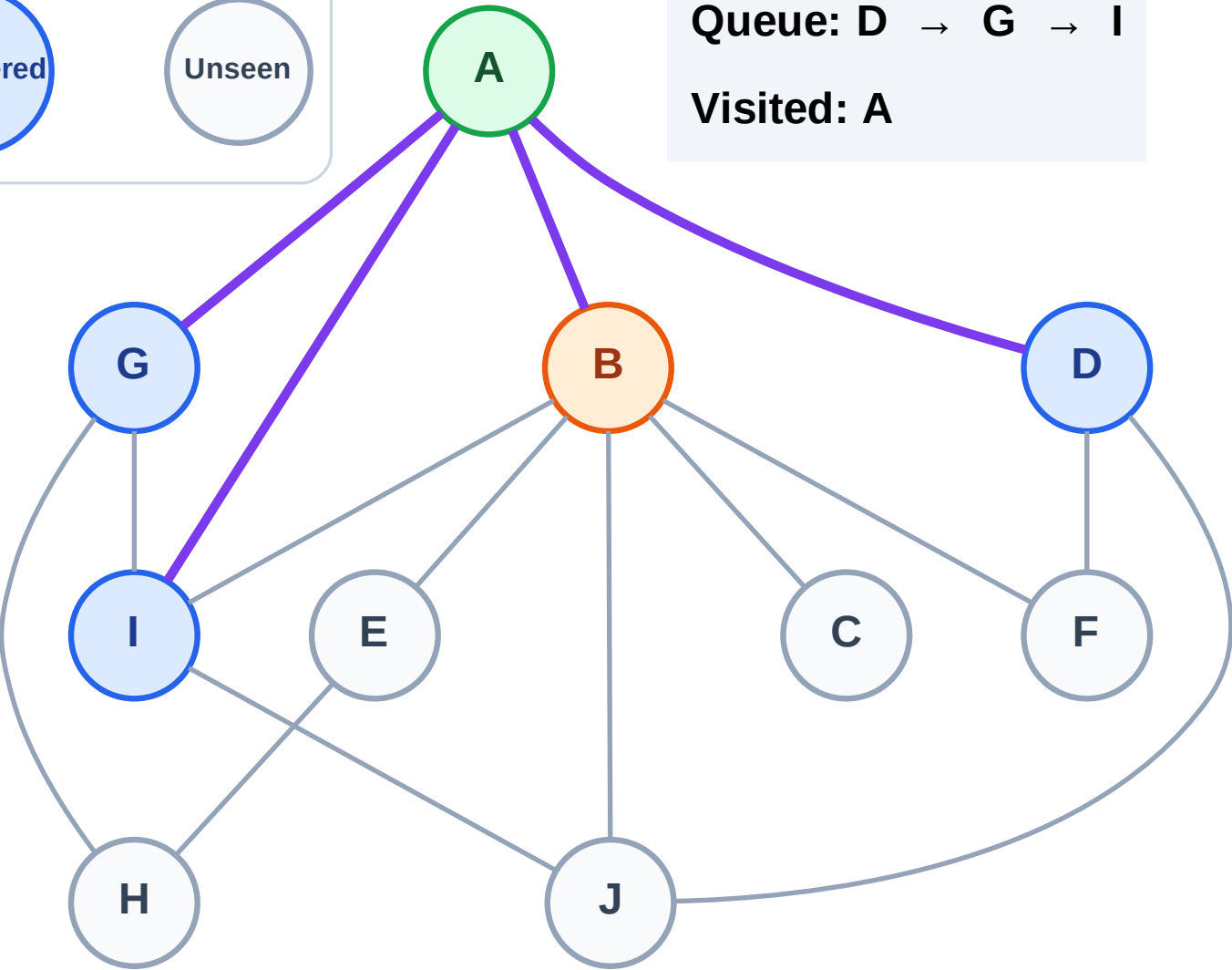
Processing

Discovered

Unseen

Queue: D → G → I

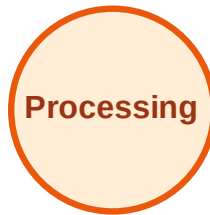
Visited: A



# BFS Traversal

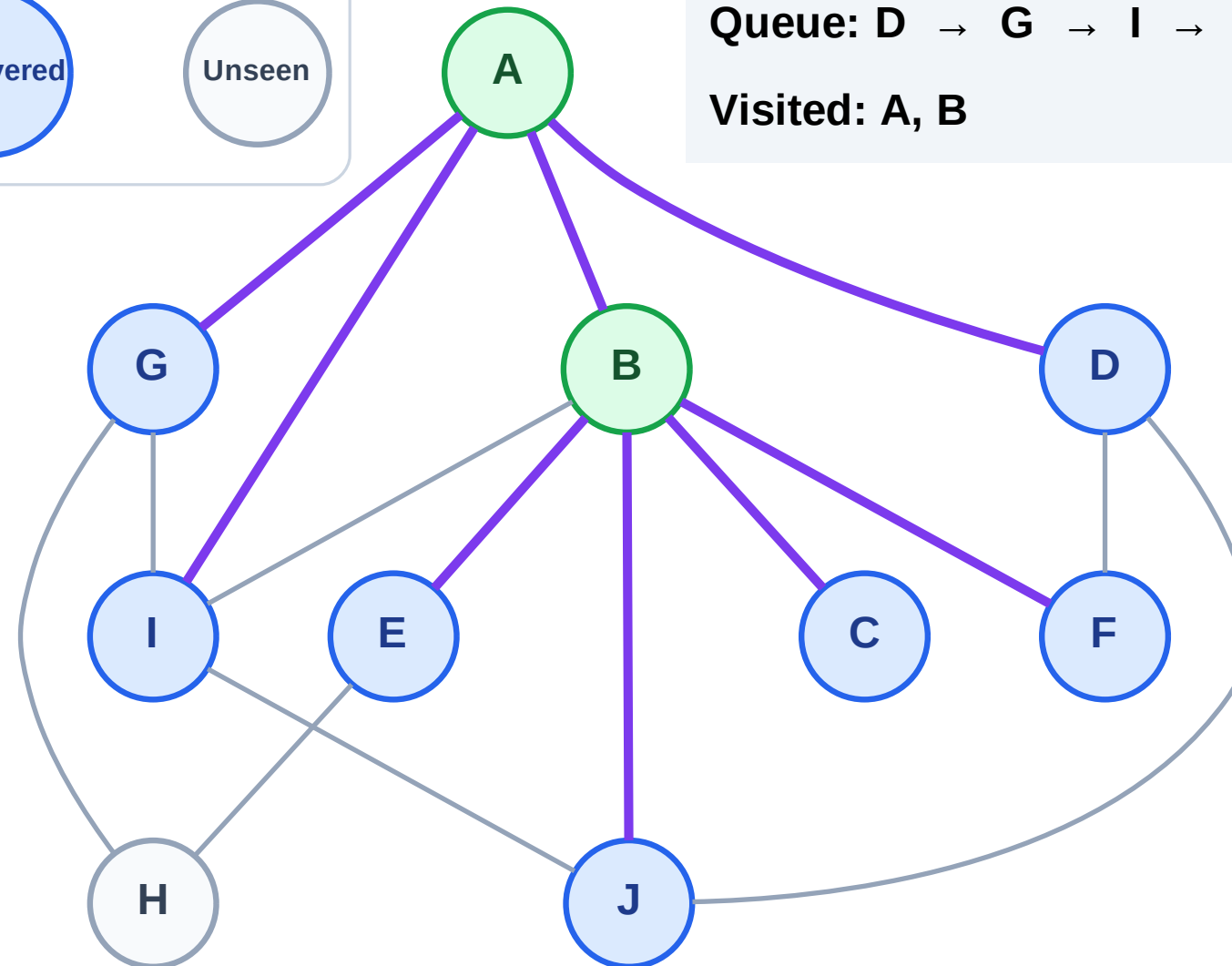
Step 5: Discover C, E, F, J from B

## Legend



Queue: D → G → I → C → E → F → J

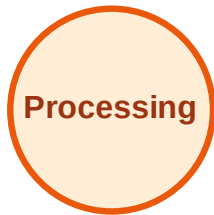
Visited: A, B



# BFS Traversal

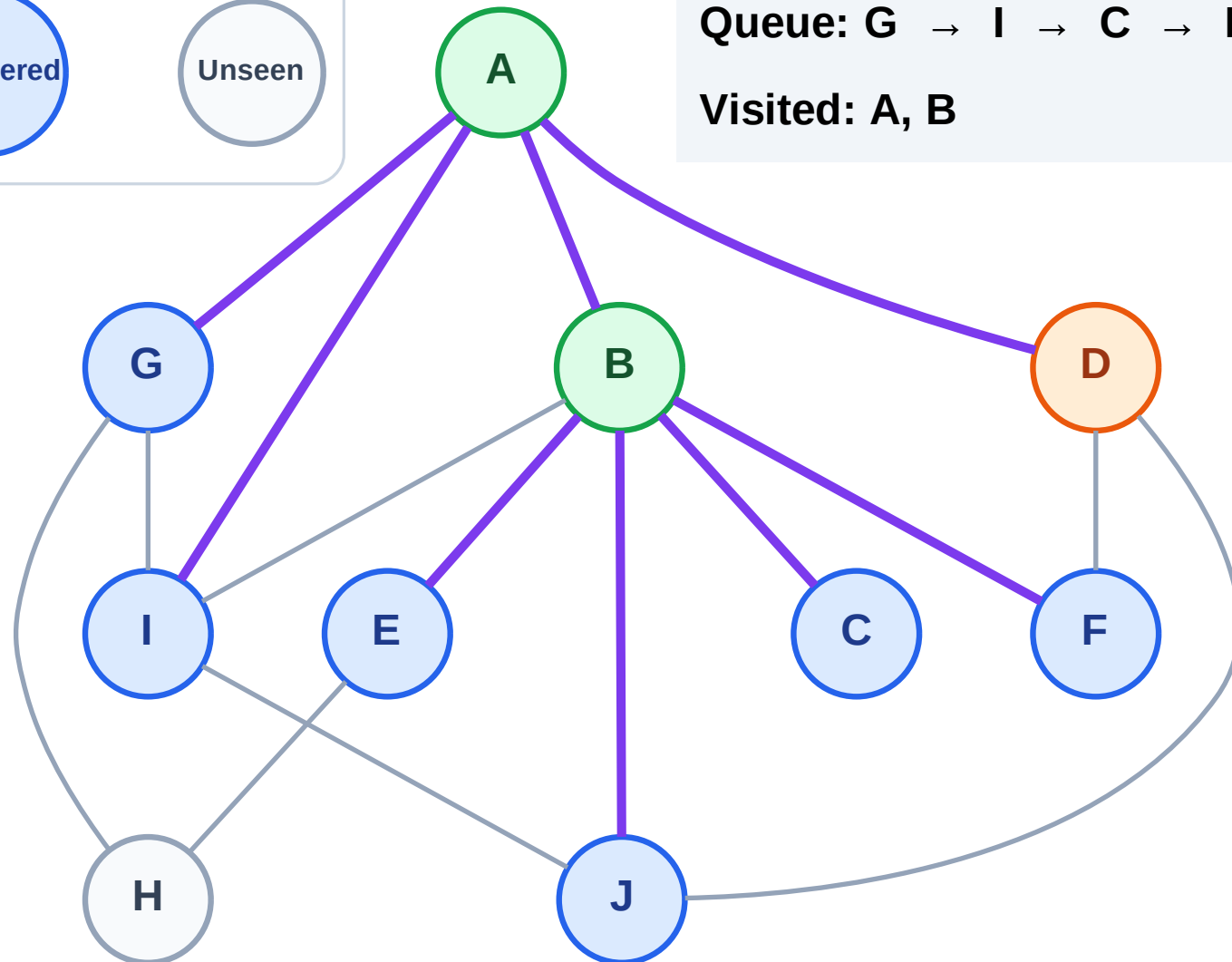
Step 6: Process node D

## Legend



Queue: G → I → C → E → F → J

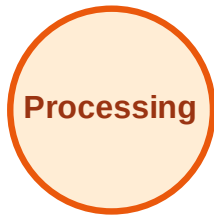
Visited: A, B



# BFS Traversal

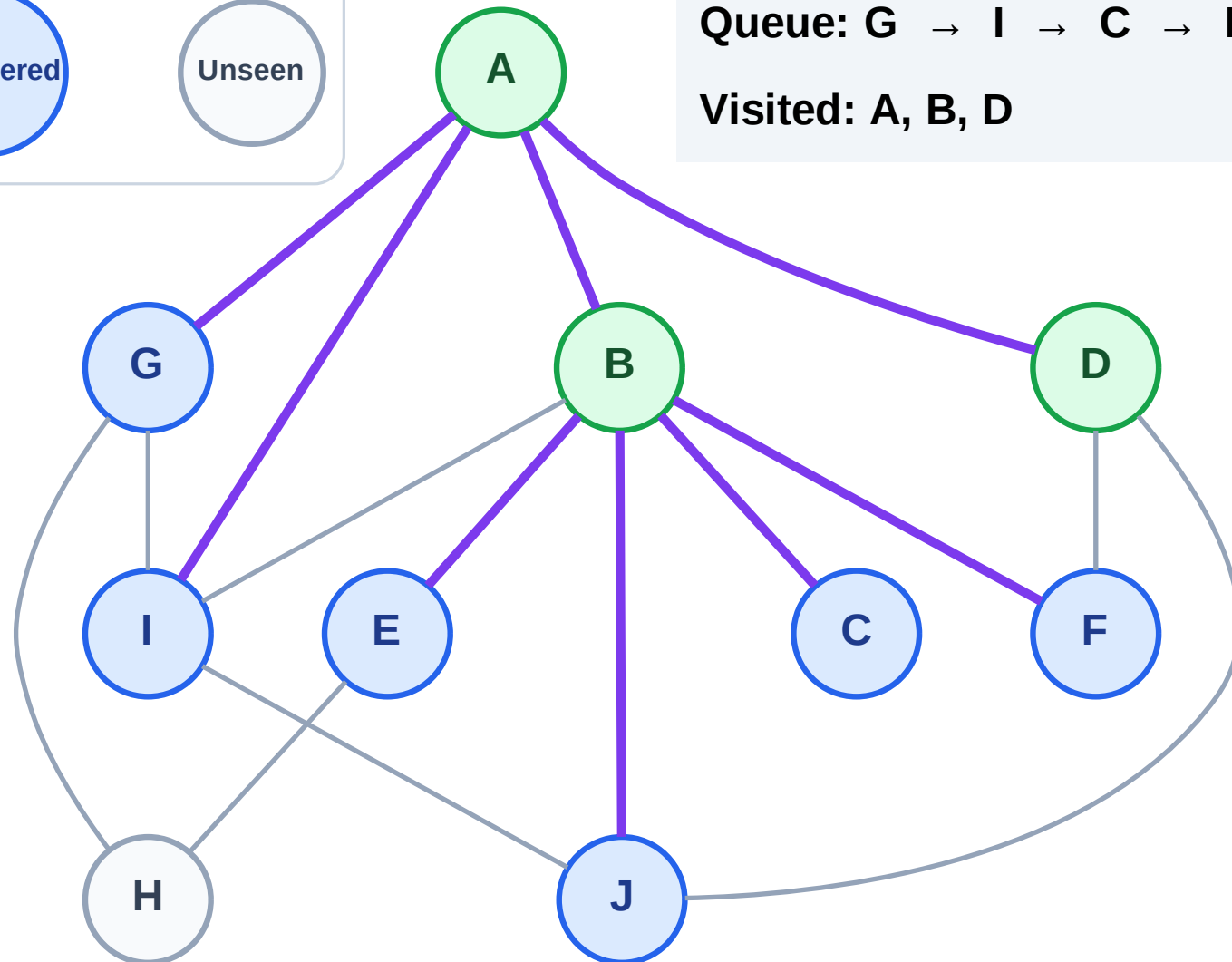
Step 7: No new neighbors from D

## Legend



Queue: G → I → C → E → F → J

Visited: A, B, D



# BFS Traversal

Step 8: Process node G

Legend

Complete

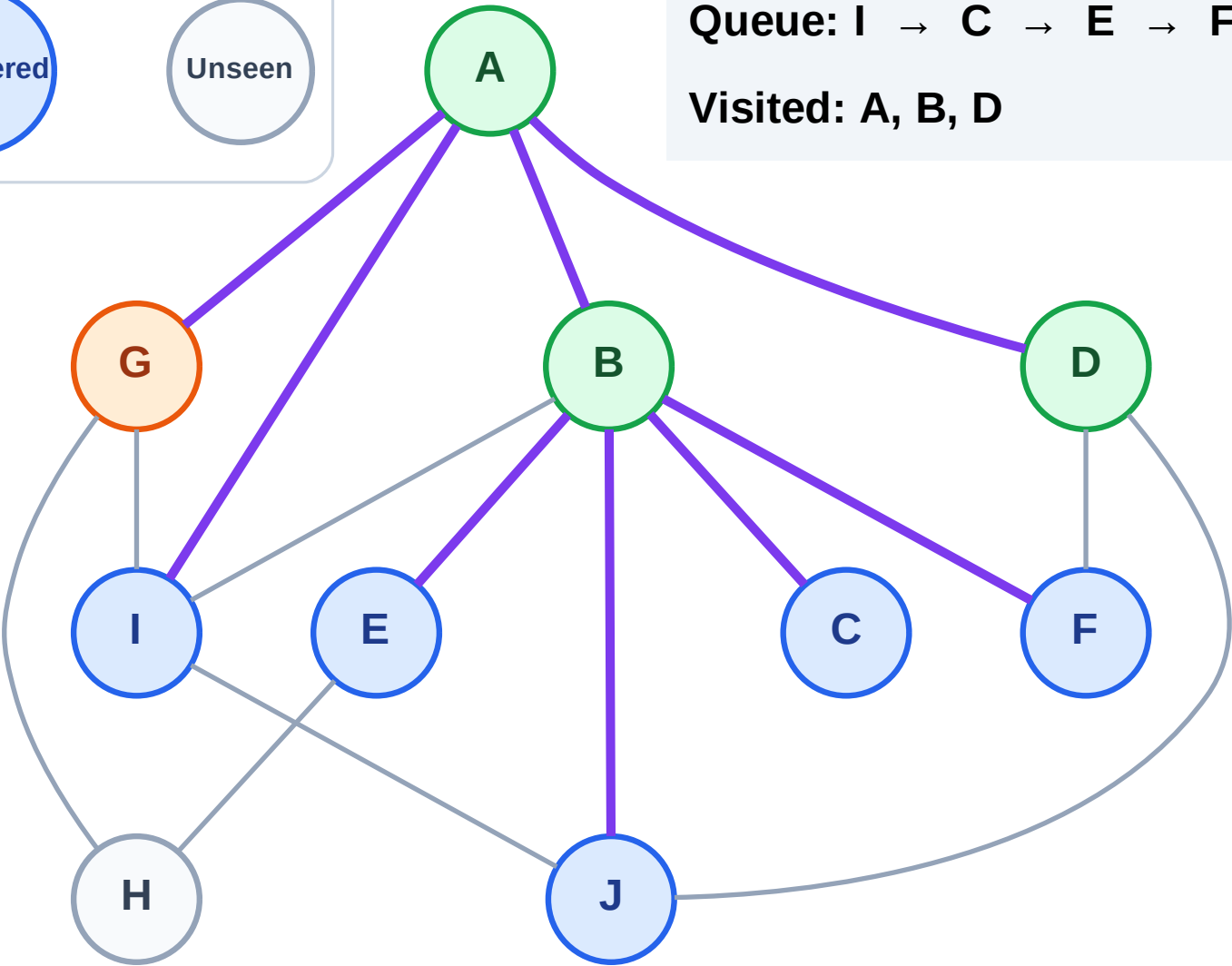
Processing

Discovered

Unseen

Queue: I → C → E → F → J

Visited: A, B, D





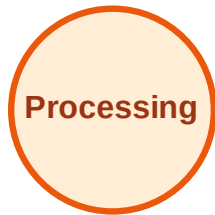
# BFS Traversal

Step 9: Discover H from G

## Legend



Complete



Processing



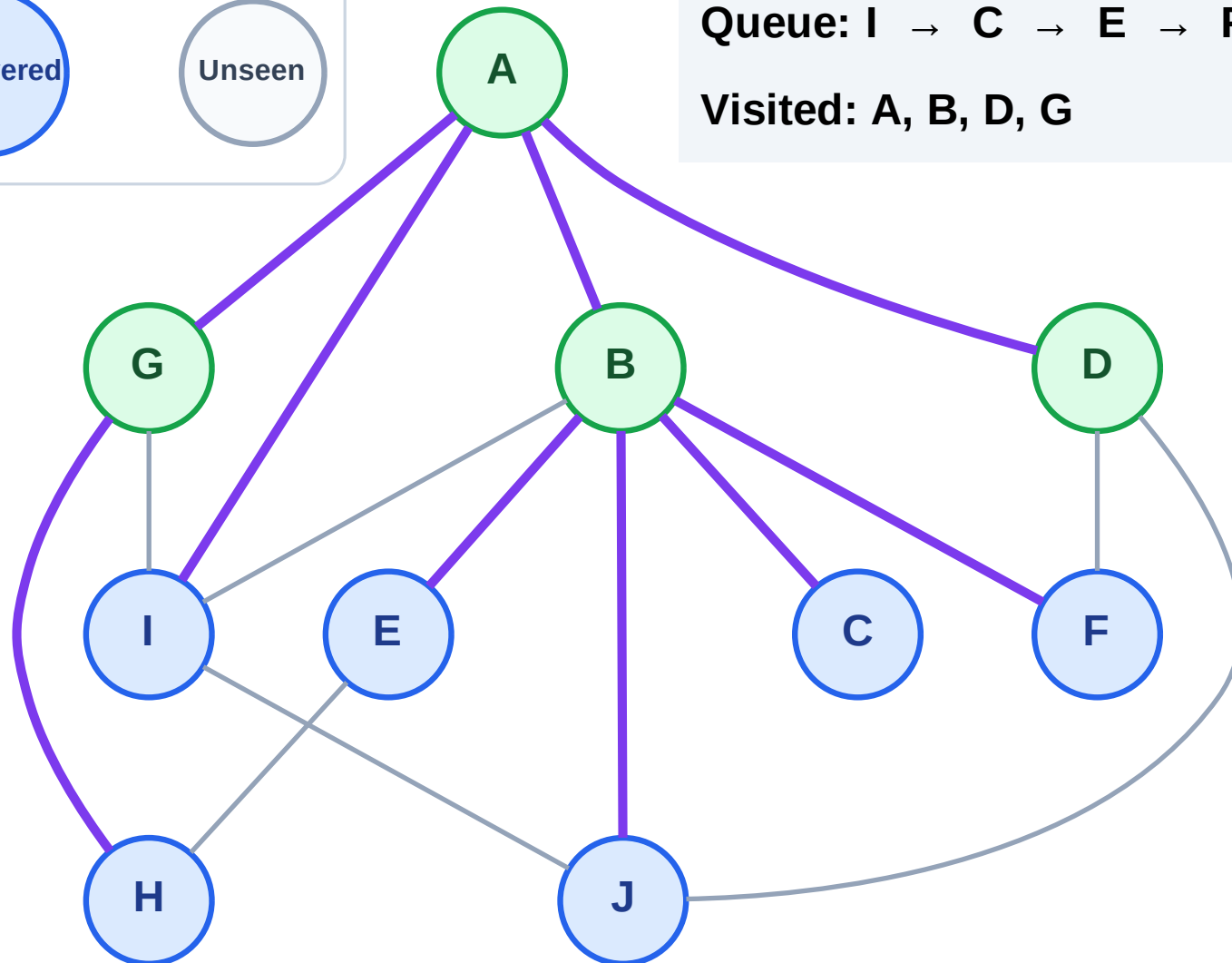
Discovered



Unseen

Queue: I → C → E → F → J → H

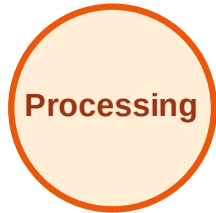
Visited: A, B, D, G



# BFS Traversal

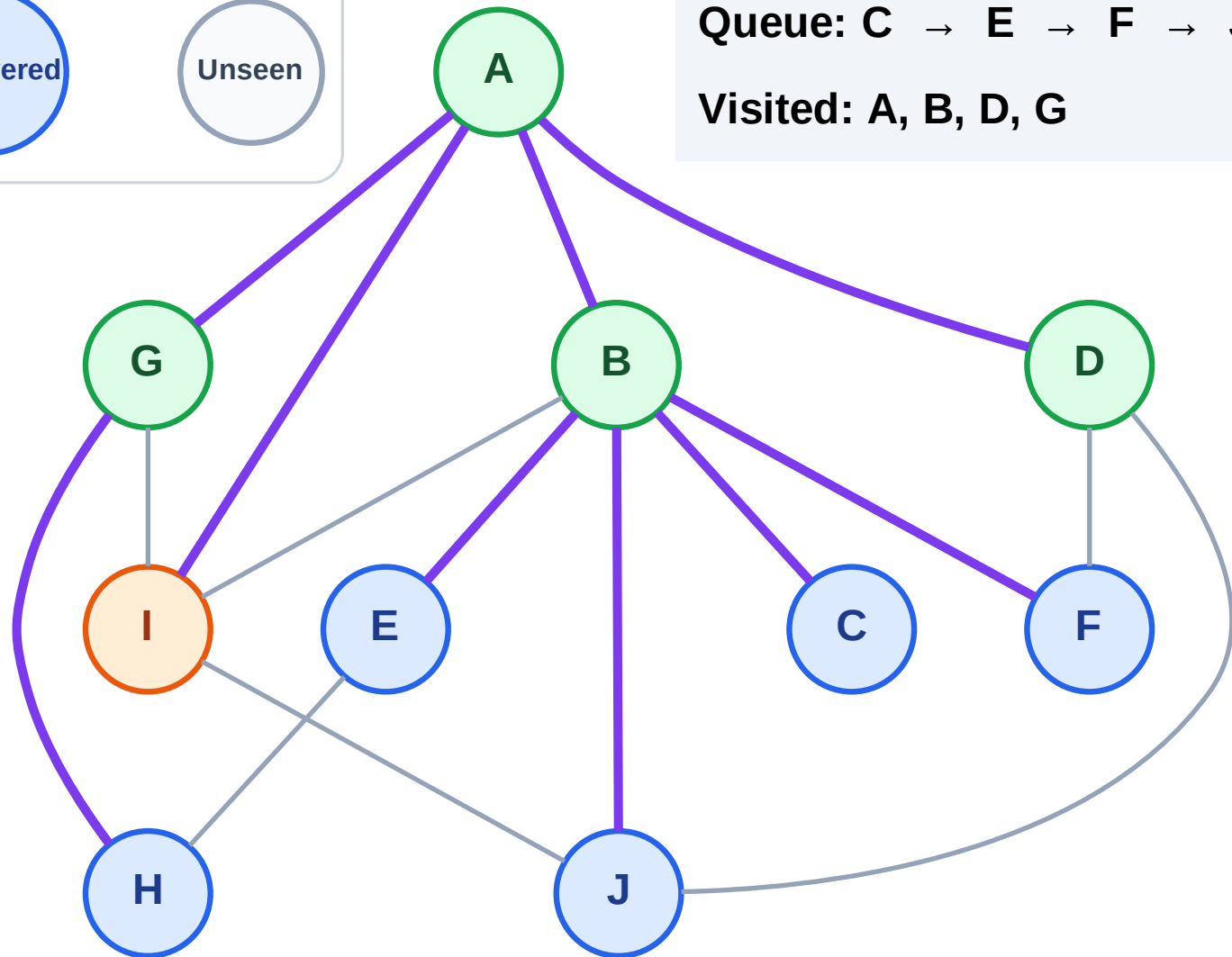
Step 10: Process node I

## Legend



Queue: C → E → F → J → H

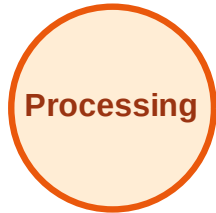
Visited: A, B, D, G



# BFS Traversal

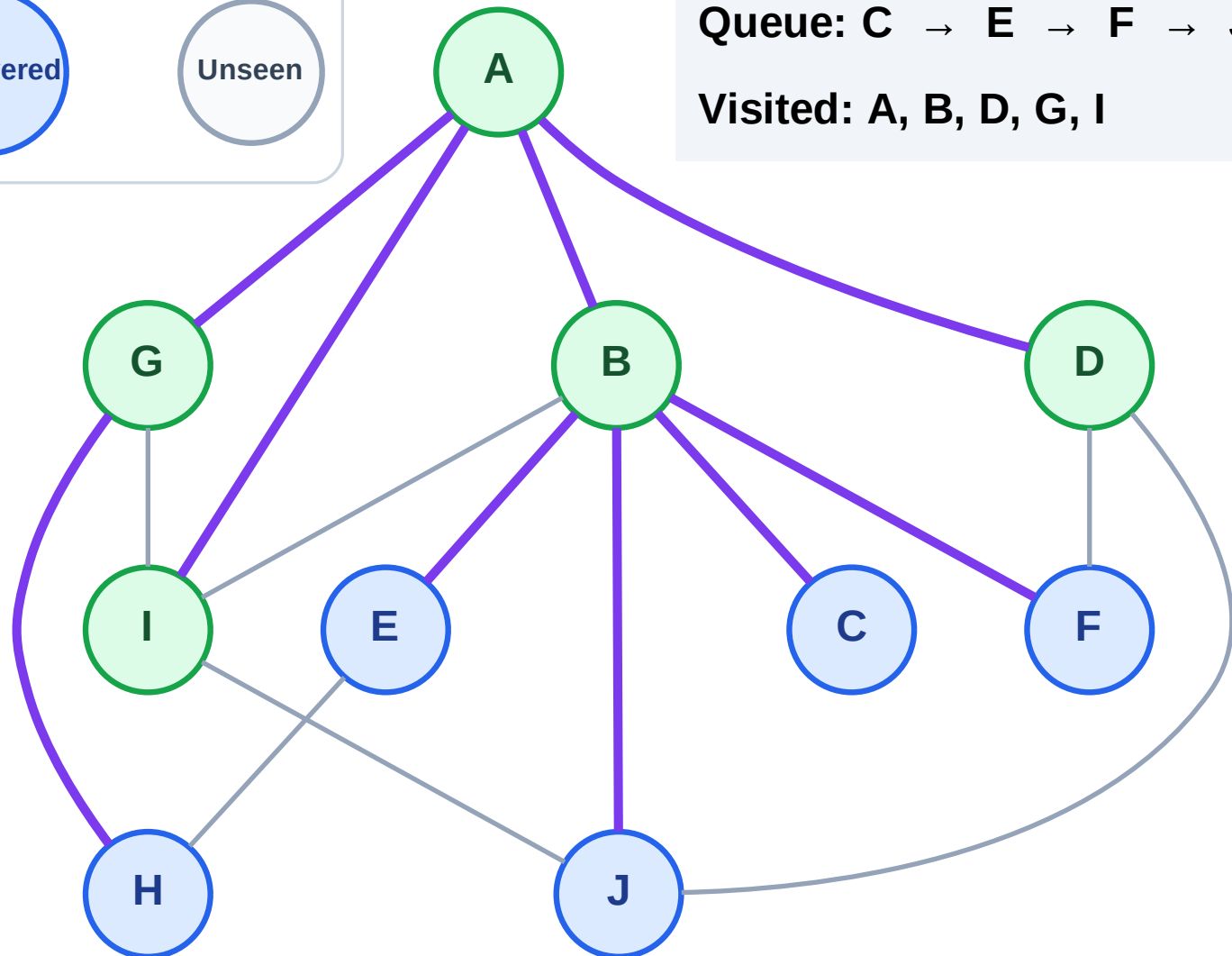
Step 11: No new neighbors from I

## Legend



Queue: C → E → F → J → H

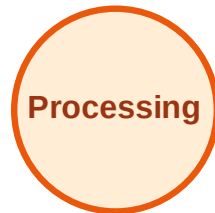
Visited: A, B, D, G, I



# BFS Traversal

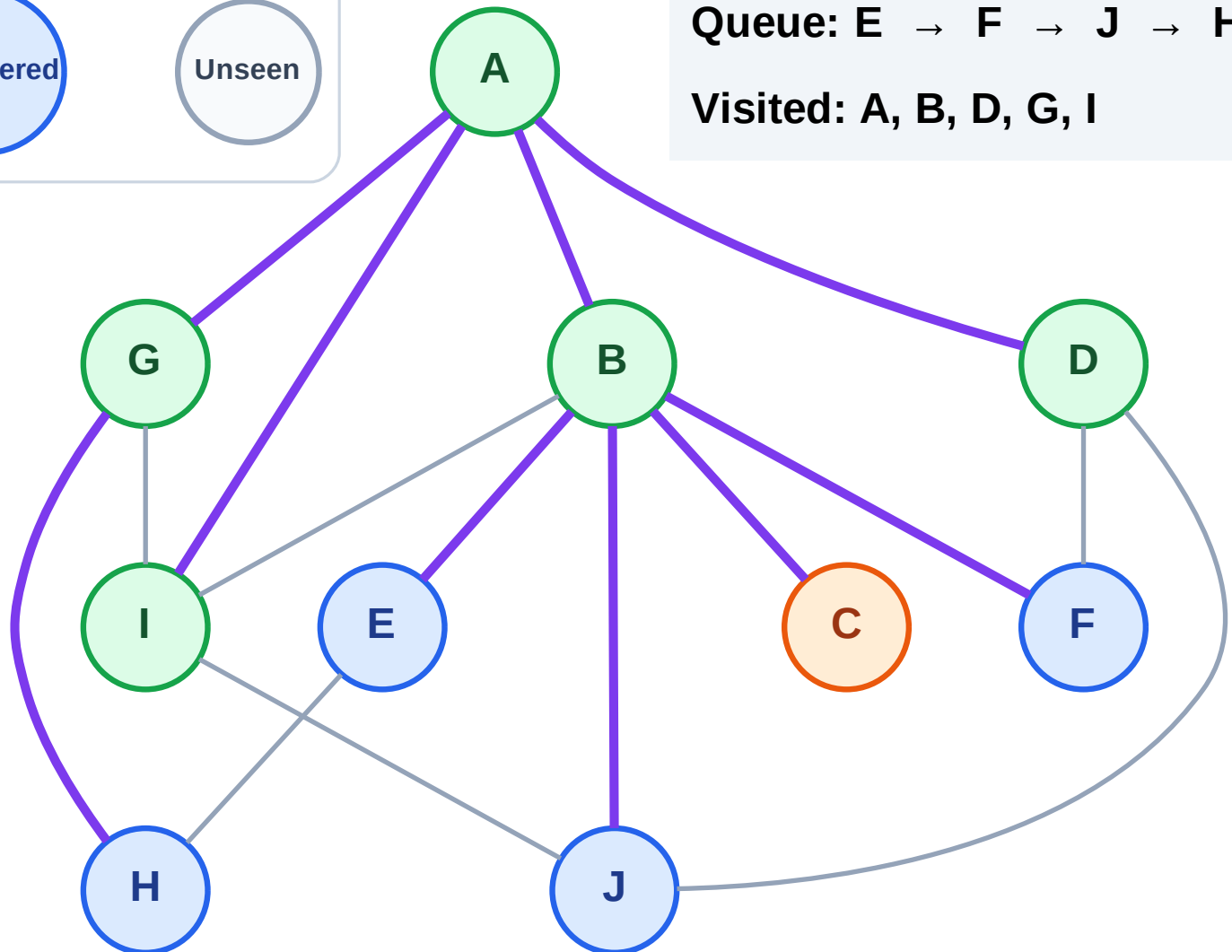
Step 12: Process node C

## Legend



Queue: E → F → J → H

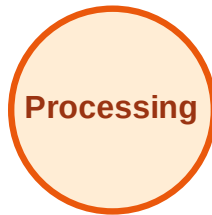
Visited: A, B, D, G, I



# BFS Traversal

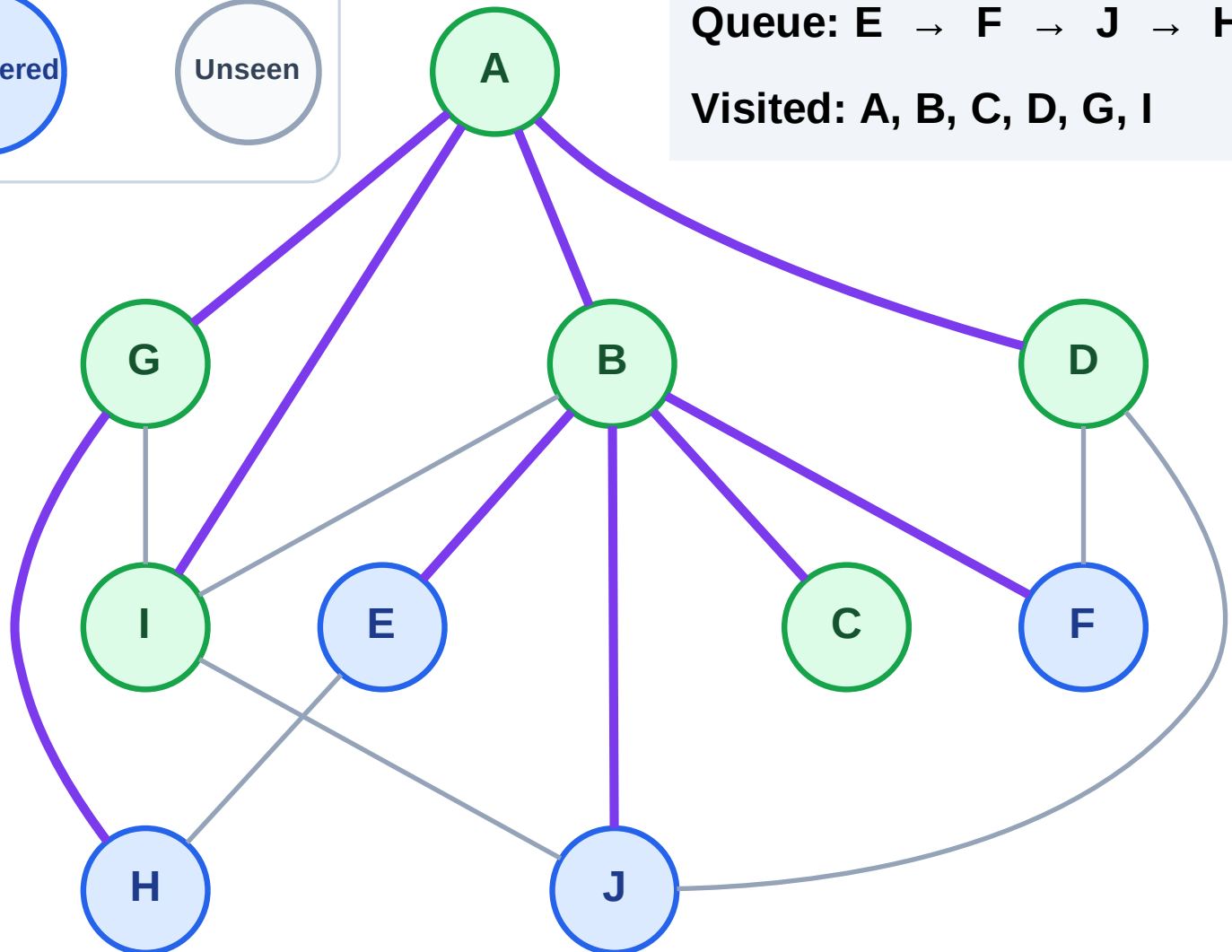
Step 13: No new neighbors from C

## Legend



Queue: E → F → J → H

Visited: A, B, C, D, G, I



# BFS Traversal

Step 14: Process node E

## Legend

Complete

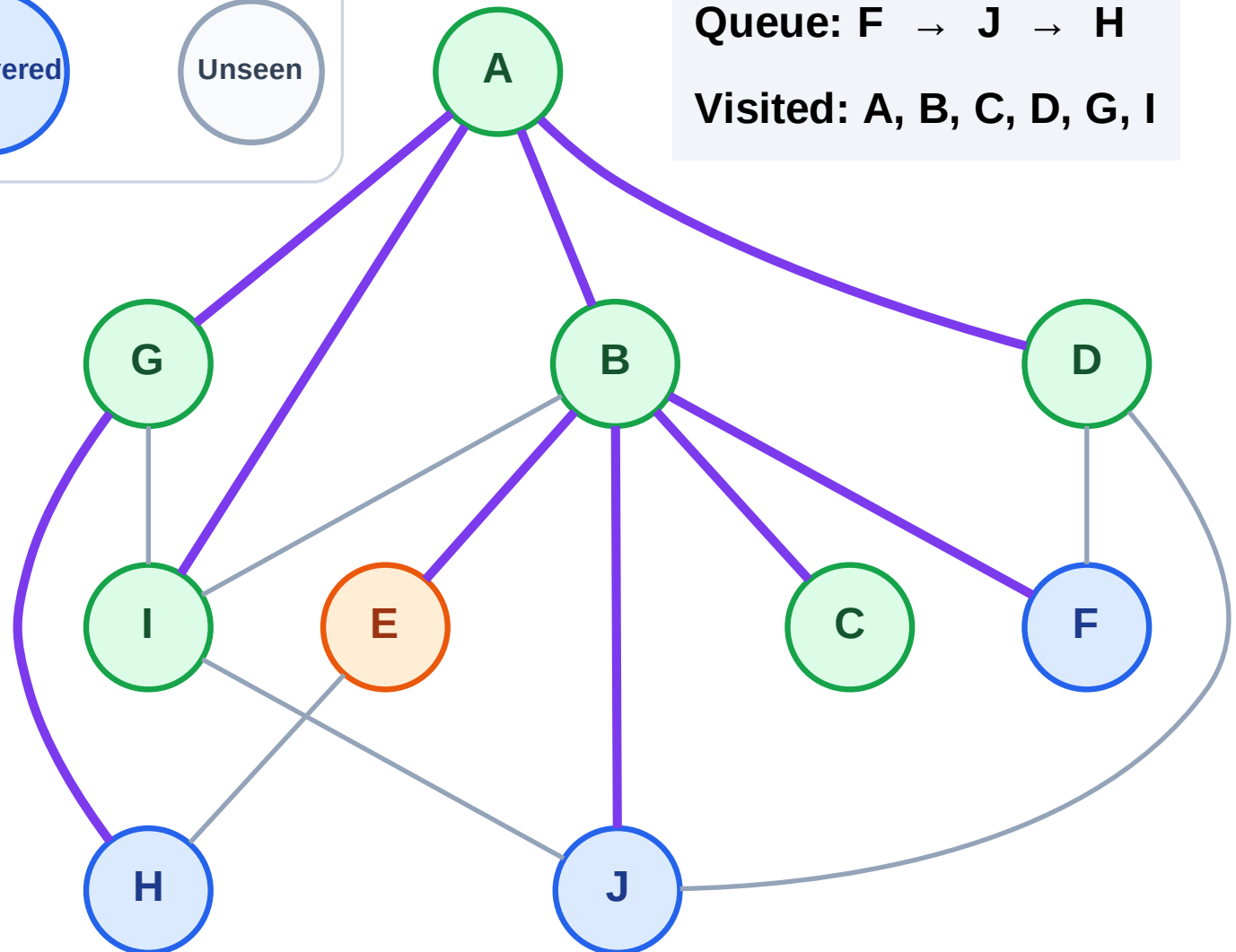
Processing

Discovered

Unseen

Queue: F → J → H

Visited: A, B, C, D, G, I



# BFS Traversal

Step 15: No new neighbors from E

## Legend

Complete

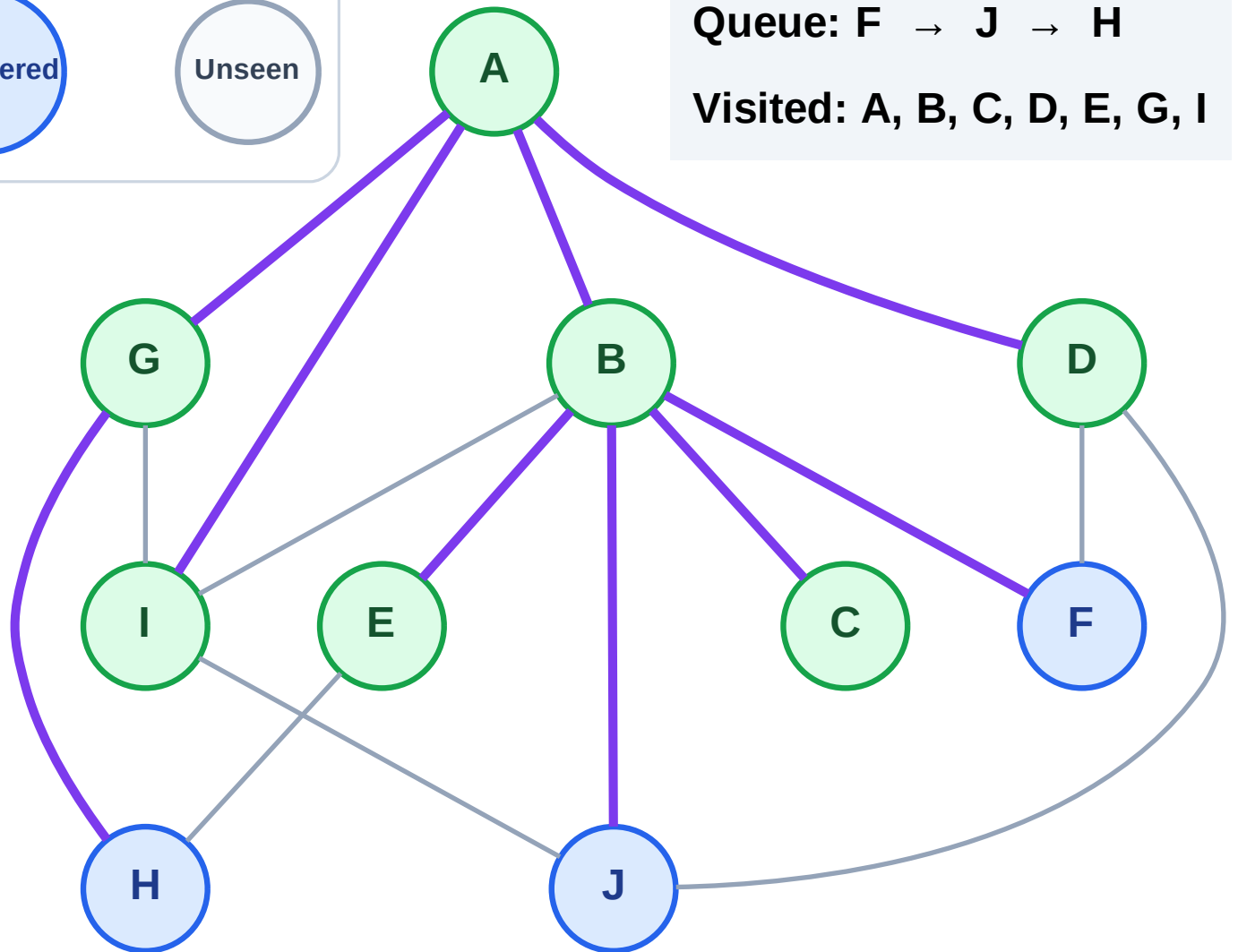
Processing

Discovered

Unseen

Queue: F → J → H

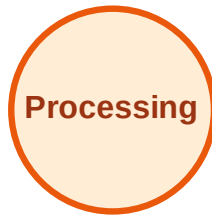
Visited: A, B, C, D, E, G, I



# BFS Traversal

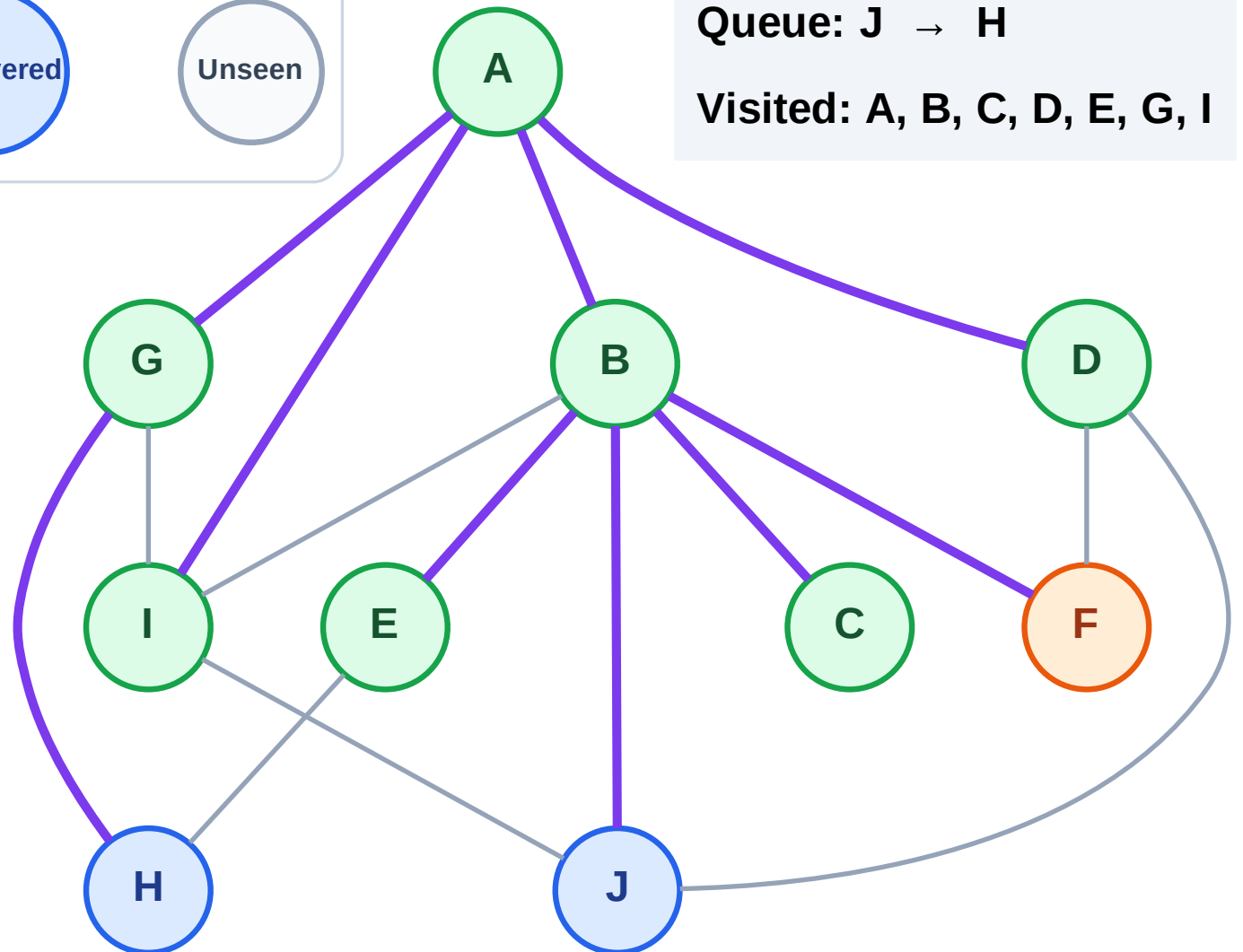
Step 16: Process node F

## Legend



Queue: J → H

Visited: A, B, C, D, E, G, I

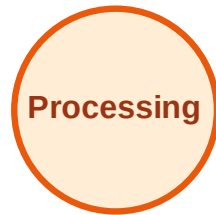




# BFS Traversal

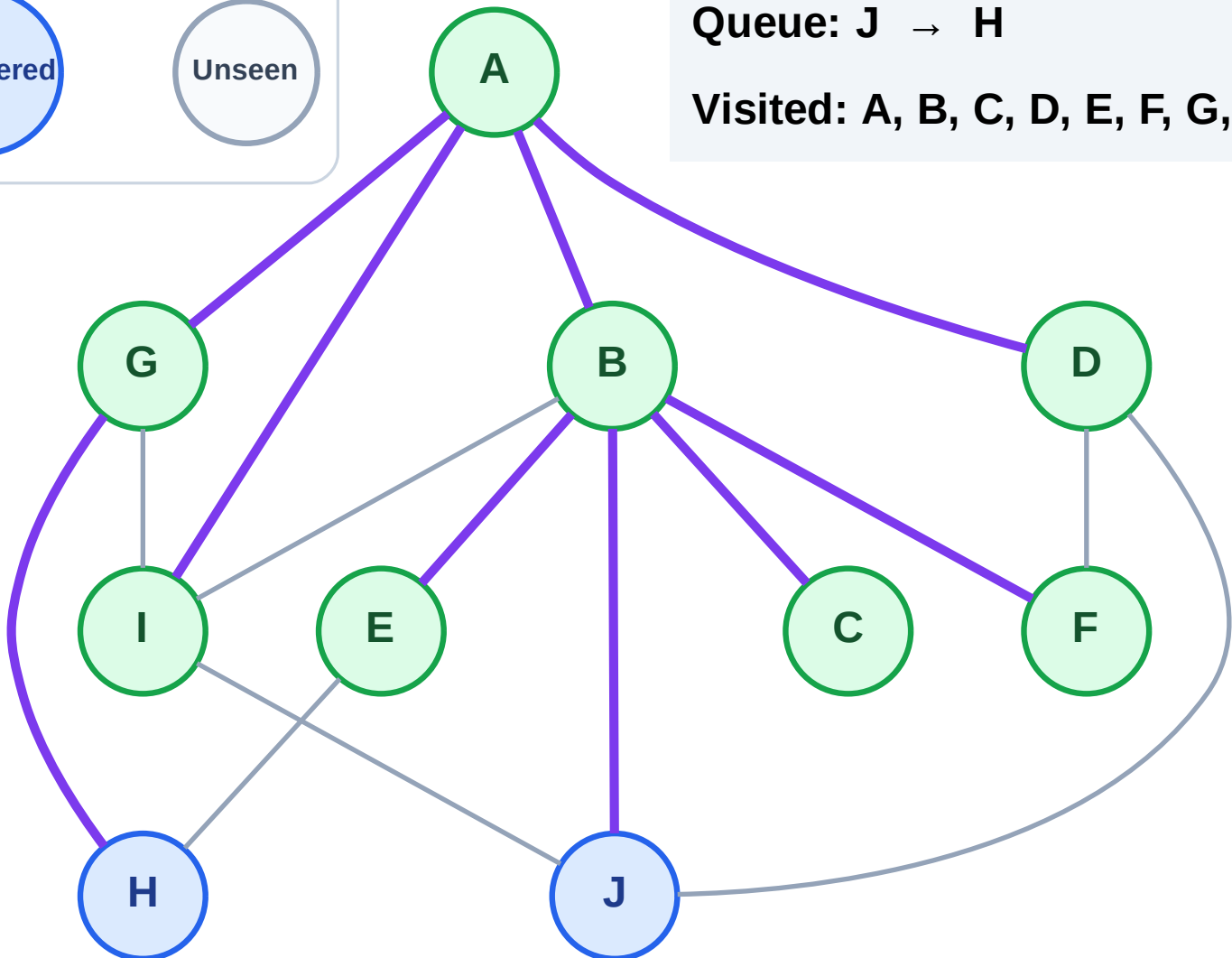
Step 17: No new neighbors from F

## Legend



Queue: J → H

Visited: A, B, C, D, E, F, G, I



# BFS Traversal

Step 18: Process node J

Legend

Complete

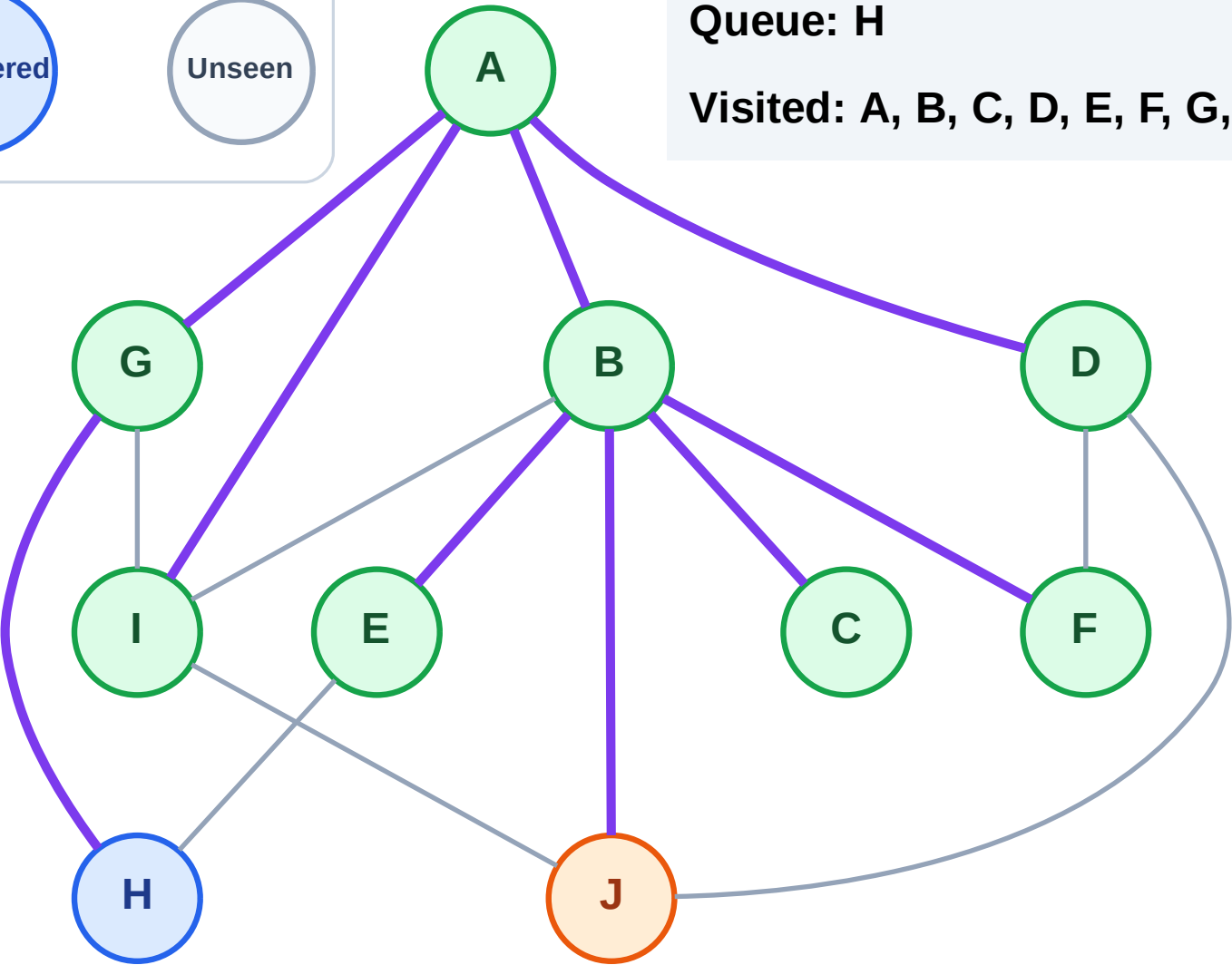
Processing

Discovered

Unseen

Queue: H

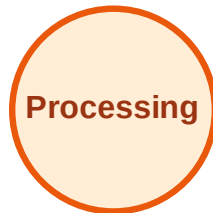
Visited: A, B, C, D, E, F, G, I



# BFS Traversal

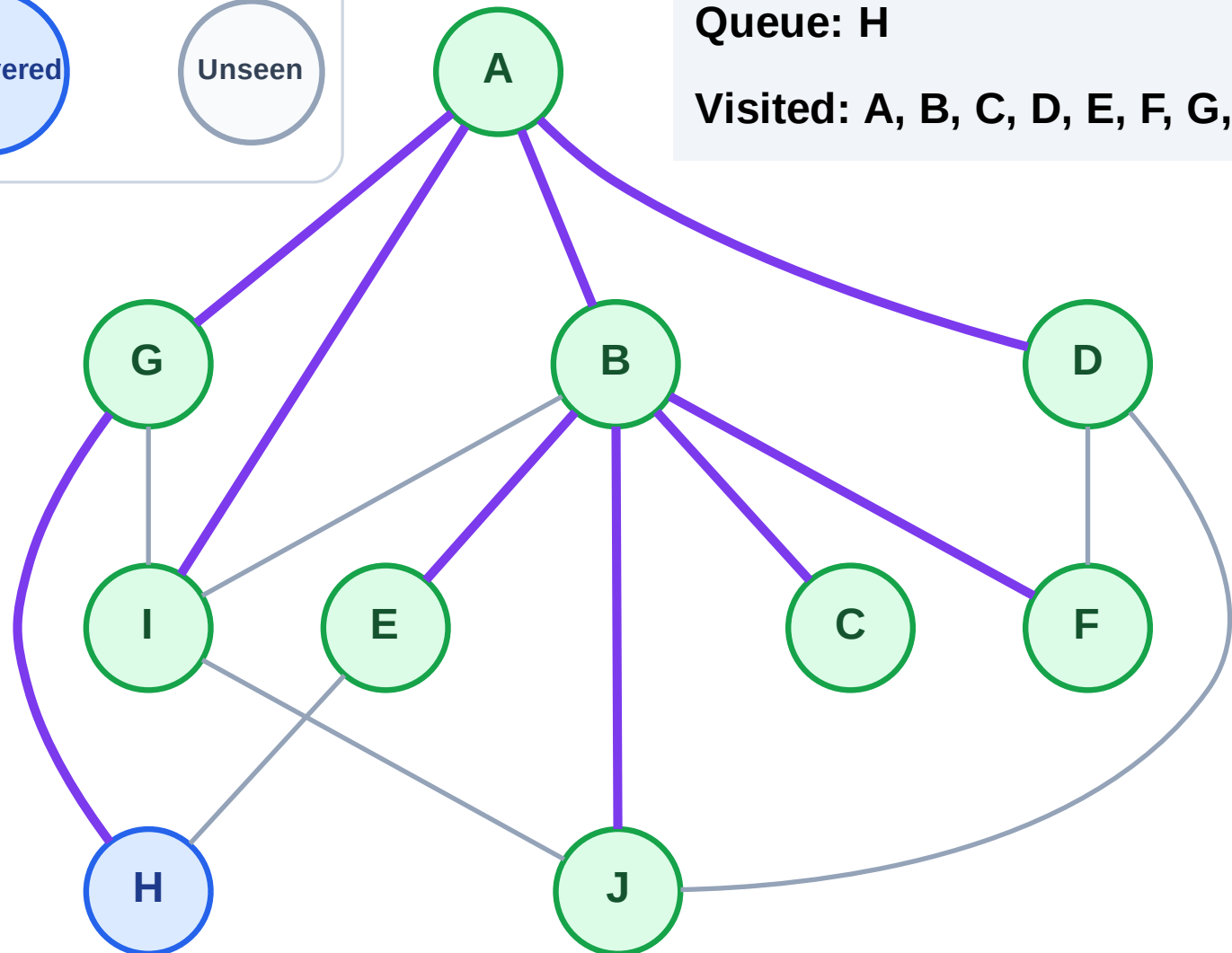
Step 19: No new neighbors from J

## Legend



Queue: H

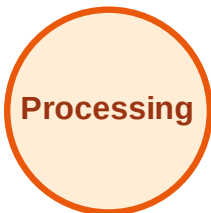
Visited: A, B, C, D, E, F, G, I, J



# BFS Traversal

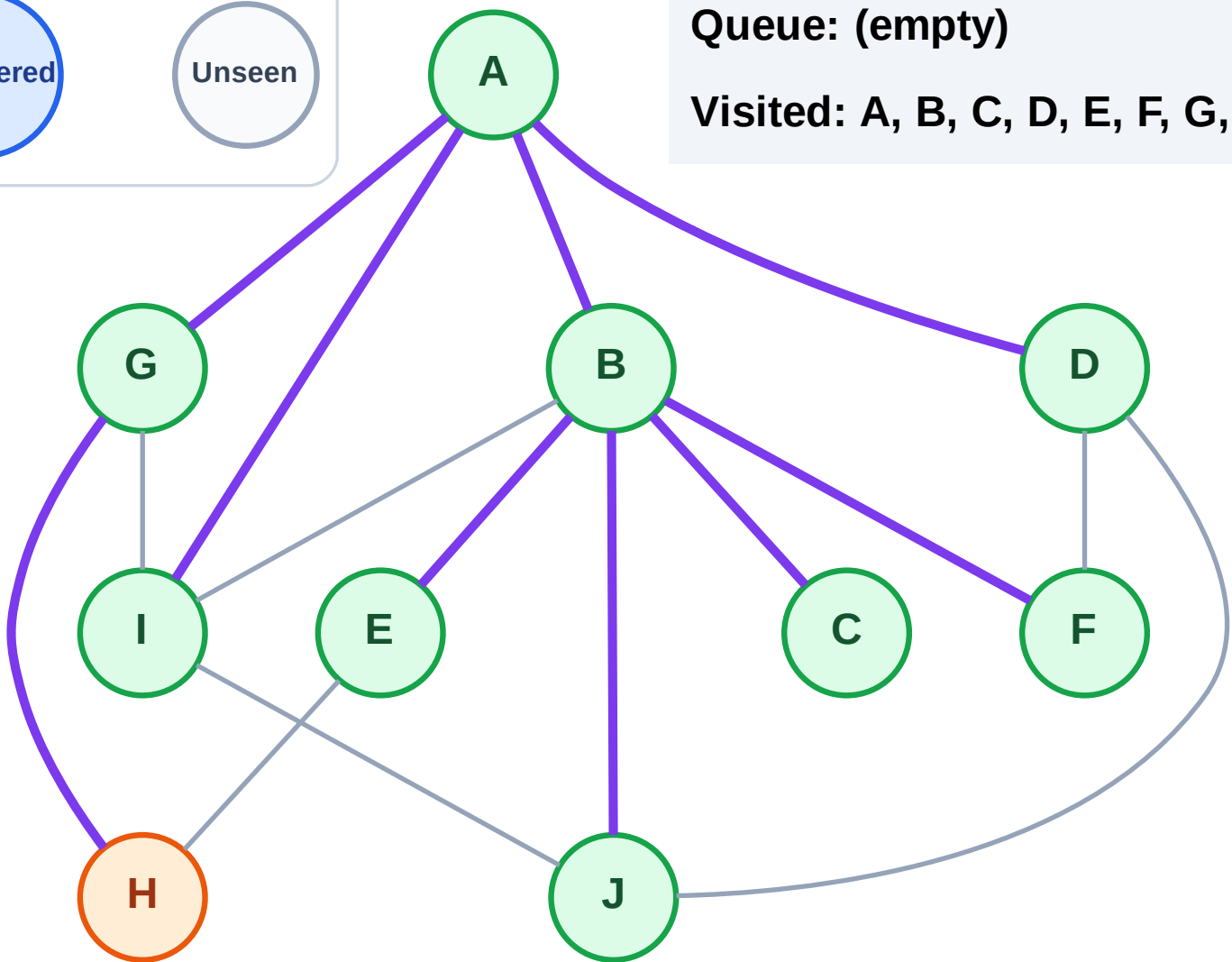
Step 20: Process node H

## Legend



Queue: (empty)

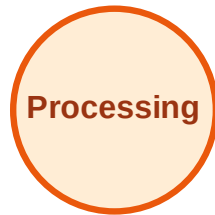
Visited: A, B, C, D, E, F, G, I, J



# BFS Traversal

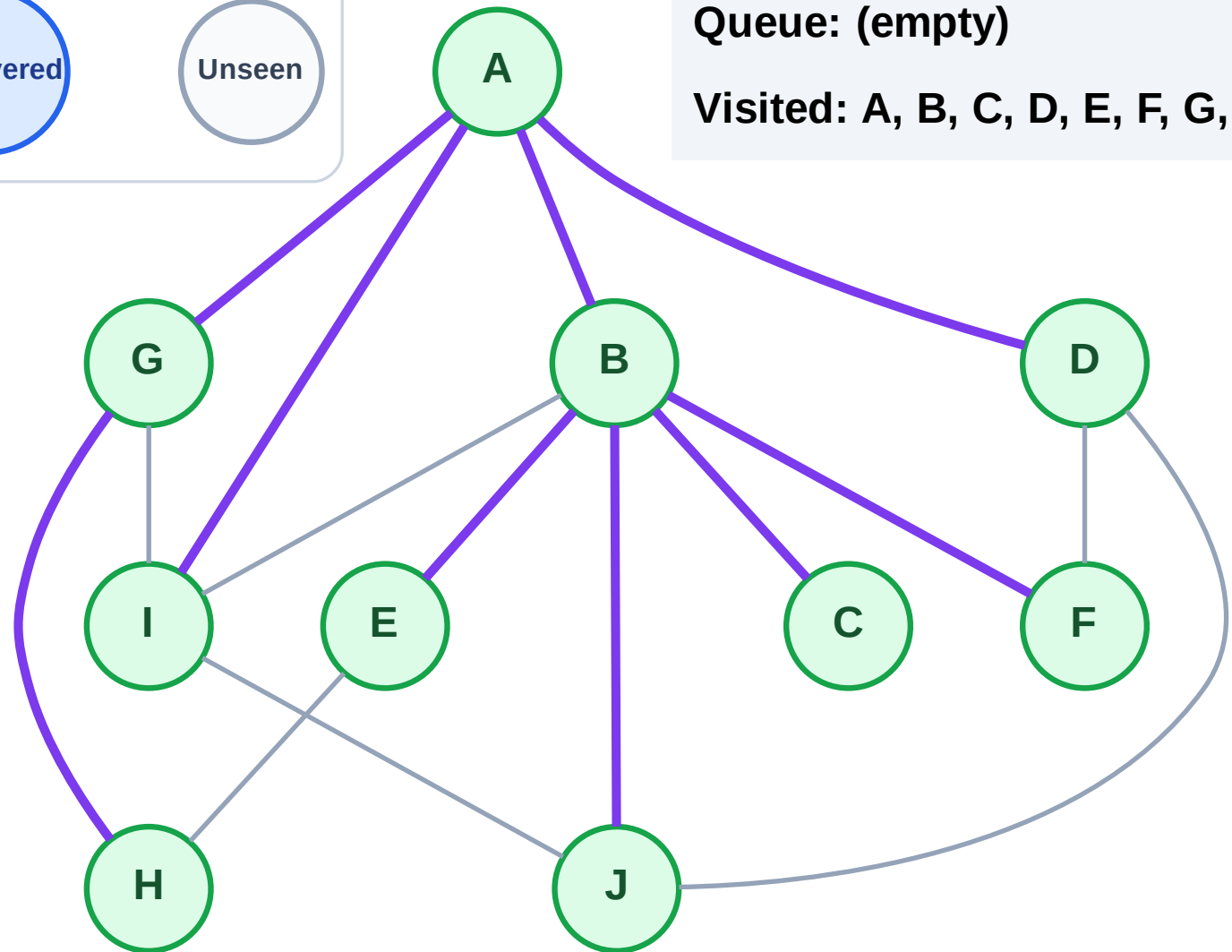
Step 21: No new neighbors from H

## Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

Step 1: Start at A

Legend

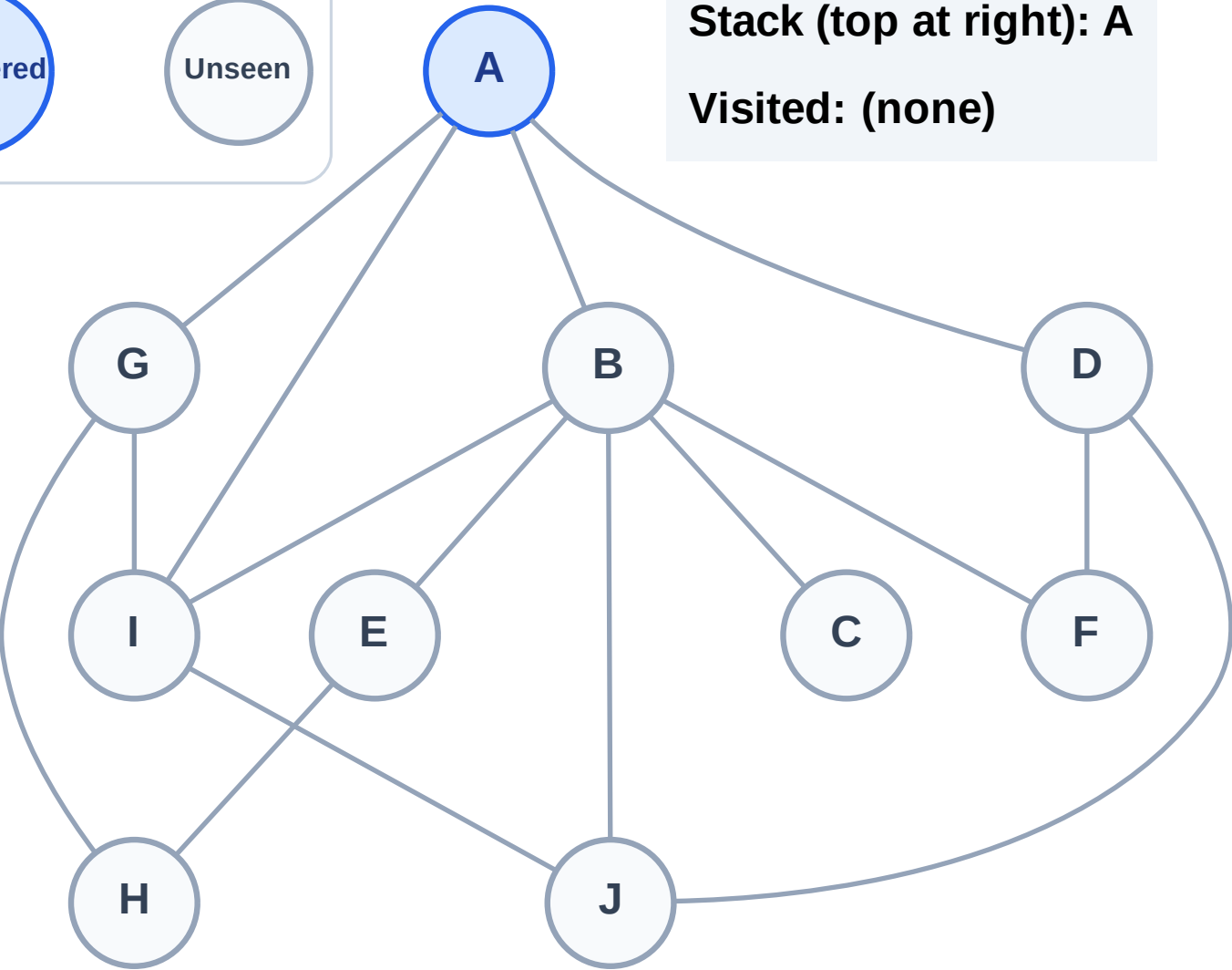
Complete

Processing

Discovered

Unseen

Stack (top at right): A  
Visited: (none)



# DFS Traversal

Step 2: Process node A

Legend

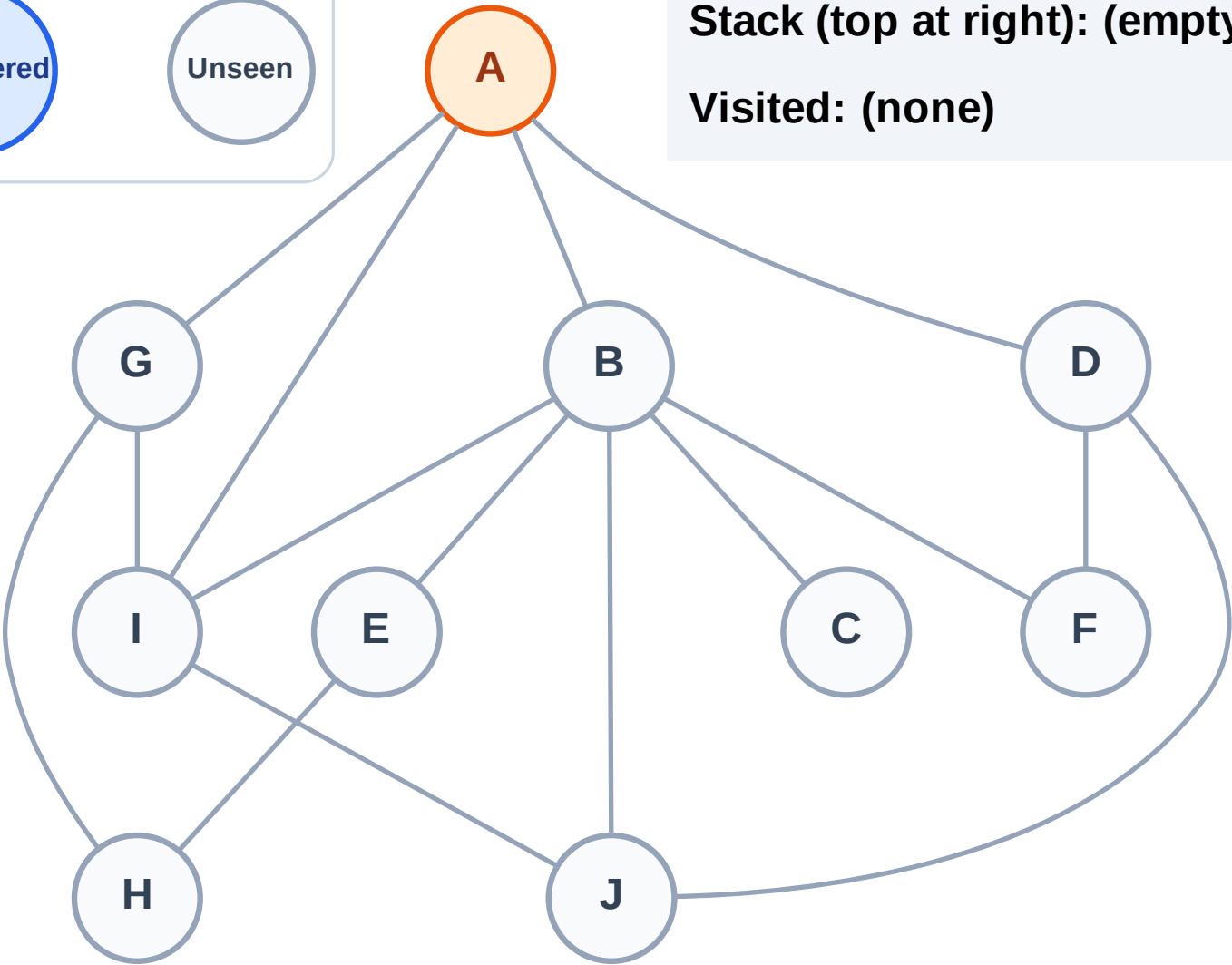
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)  
Visited: (none)



# DFS Traversal

Step 3: Discover I, G, D, B from A

Legend

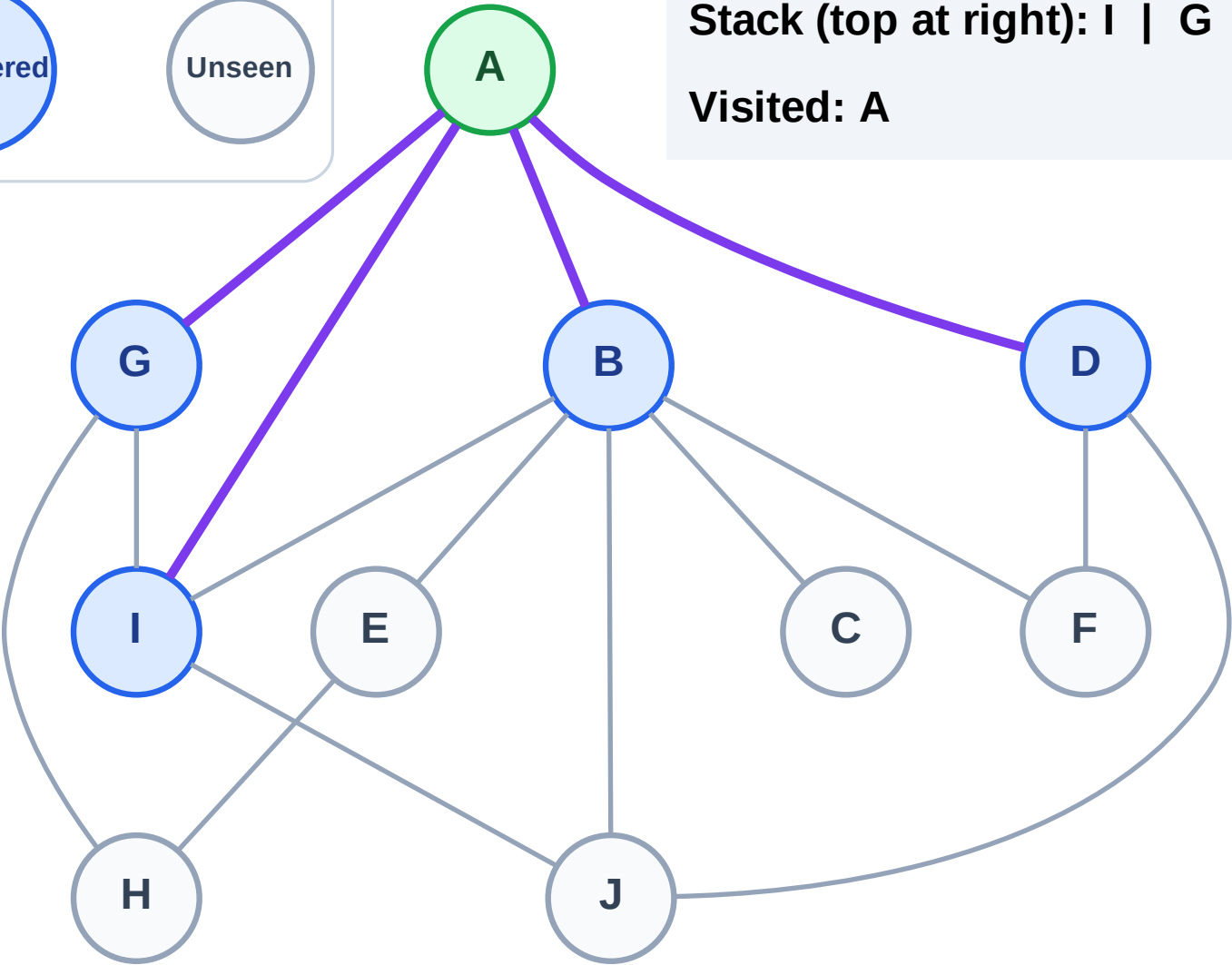
Complete

Processing

Discovered

Unseen

Stack (top at right): I | G | D | B  
Visited: A

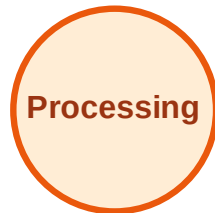




# DFS Traversal

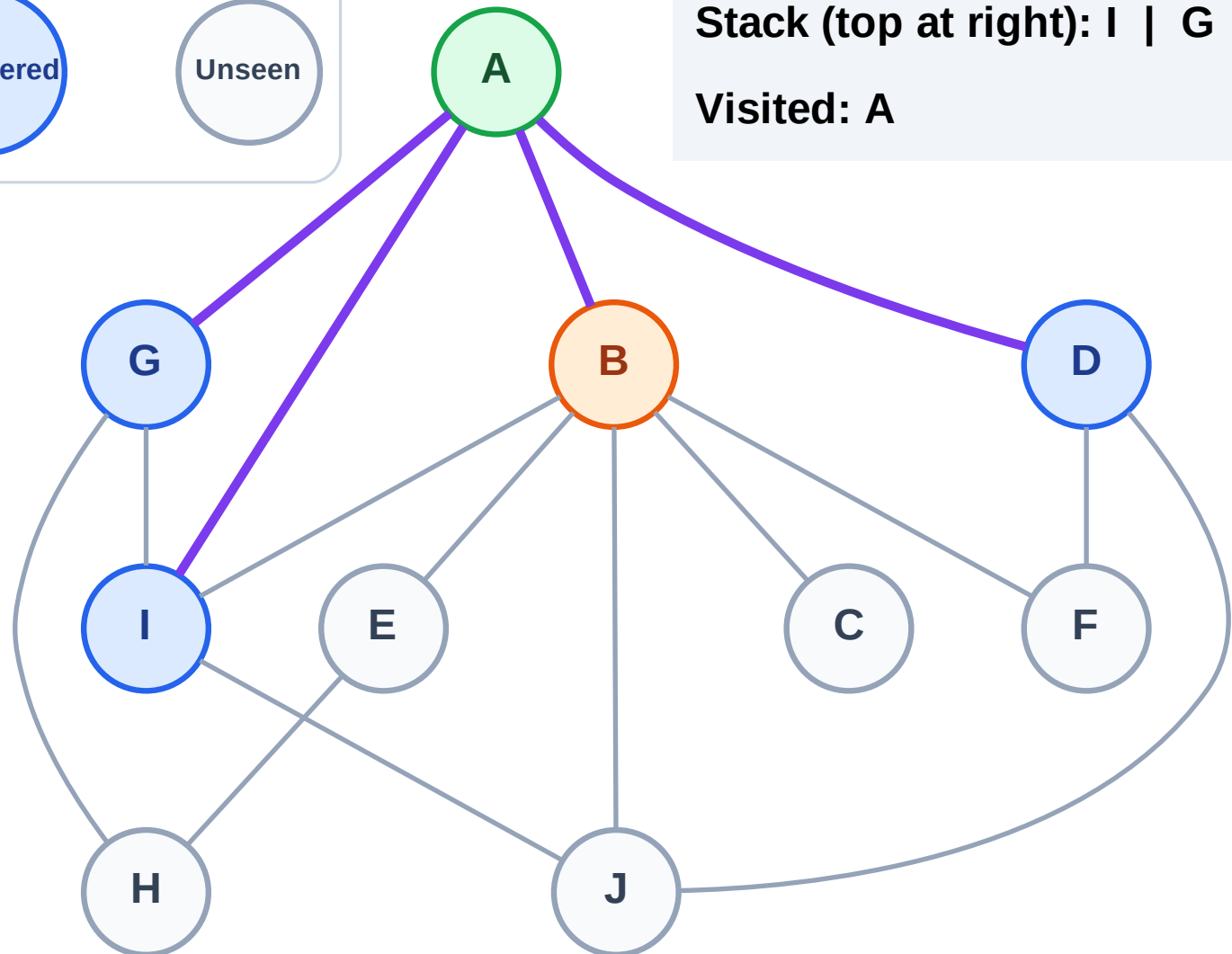
Step 4: Process node B

## Legend



Stack (top at right): I | G | D

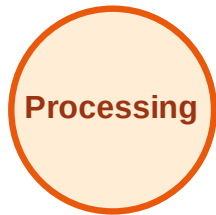
Visited: A



# DFS Traversal

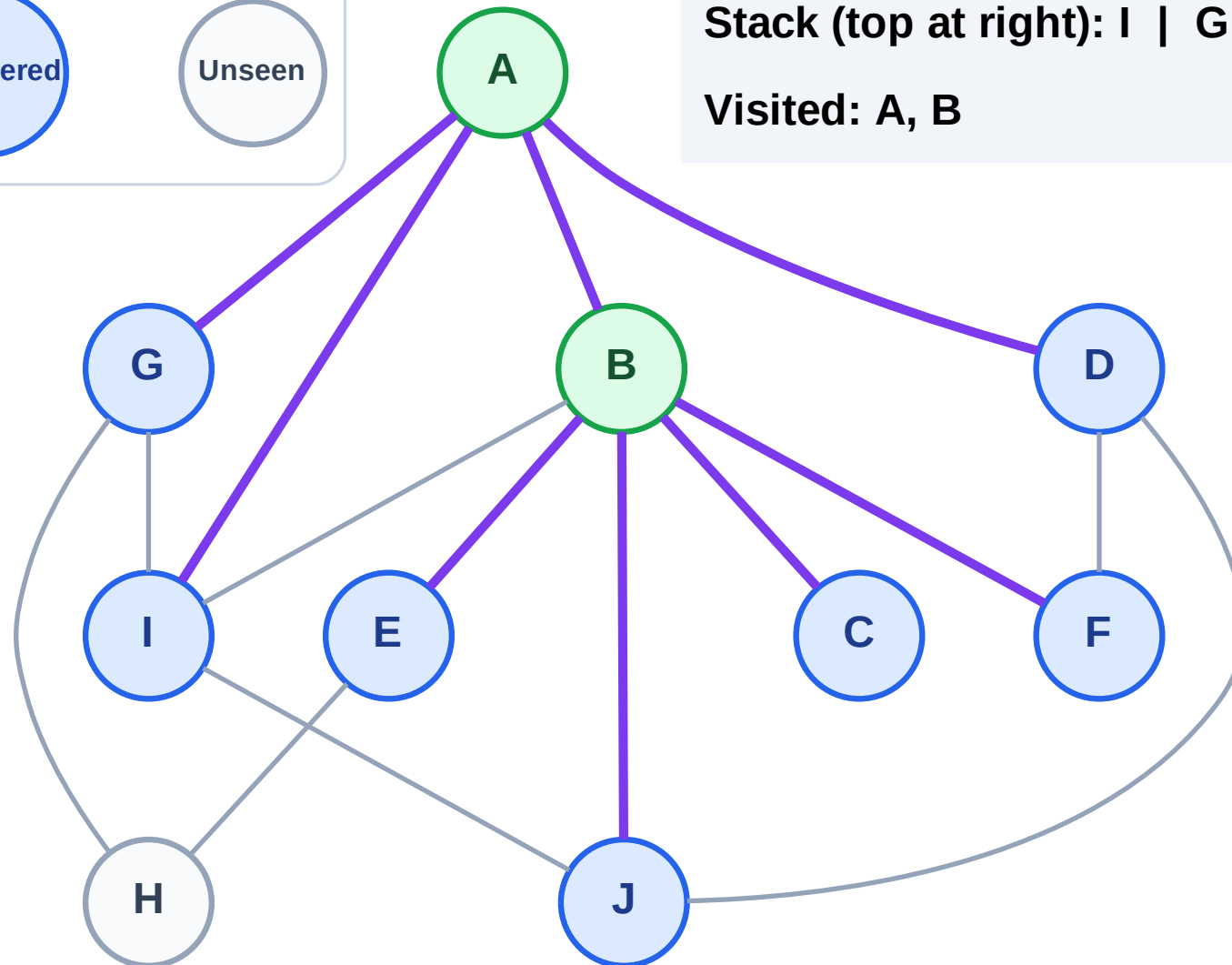
Step 5: Discover J, F, E, C from B

## Legend



Stack (top at right): I | G | D | J | F | E | C

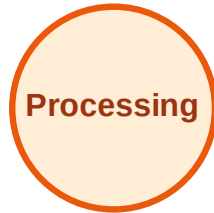
Visited: A, B



# DFS Traversal

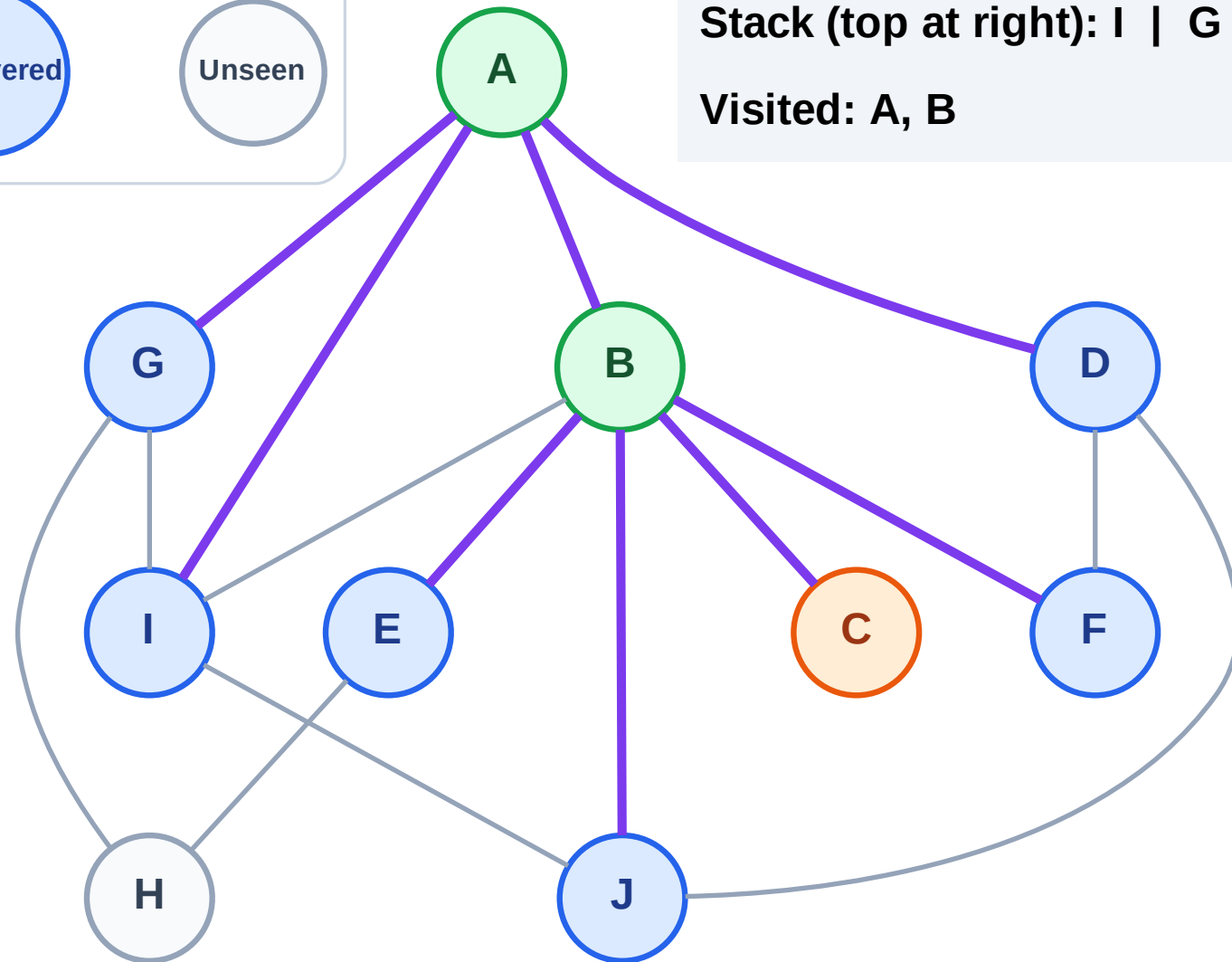
Step 6: Process node C

## Legend



Stack (top at right): I | G | D | J | F | E

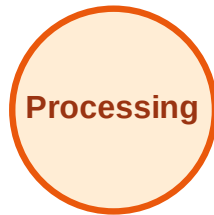
Visited: A, B



# DFS Traversal

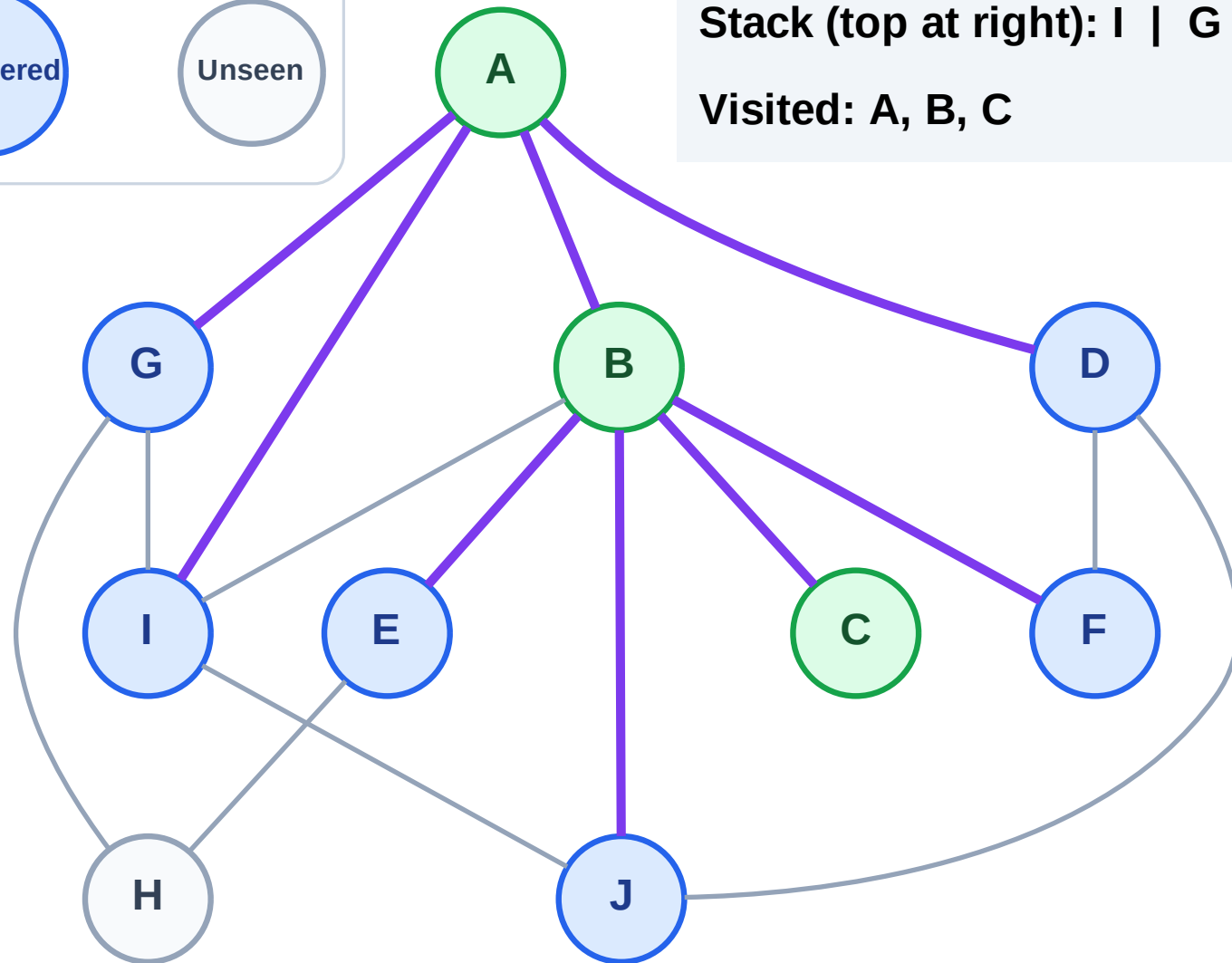
Step 7: No new neighbors from C

## Legend



Stack (top at right): I | G | D | J | F | E

Visited: A, B, C



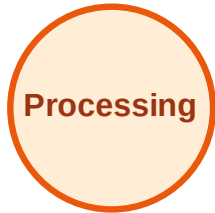
# DFS Traversal

Step 8: Process node E

## Legend



Complete



Processing



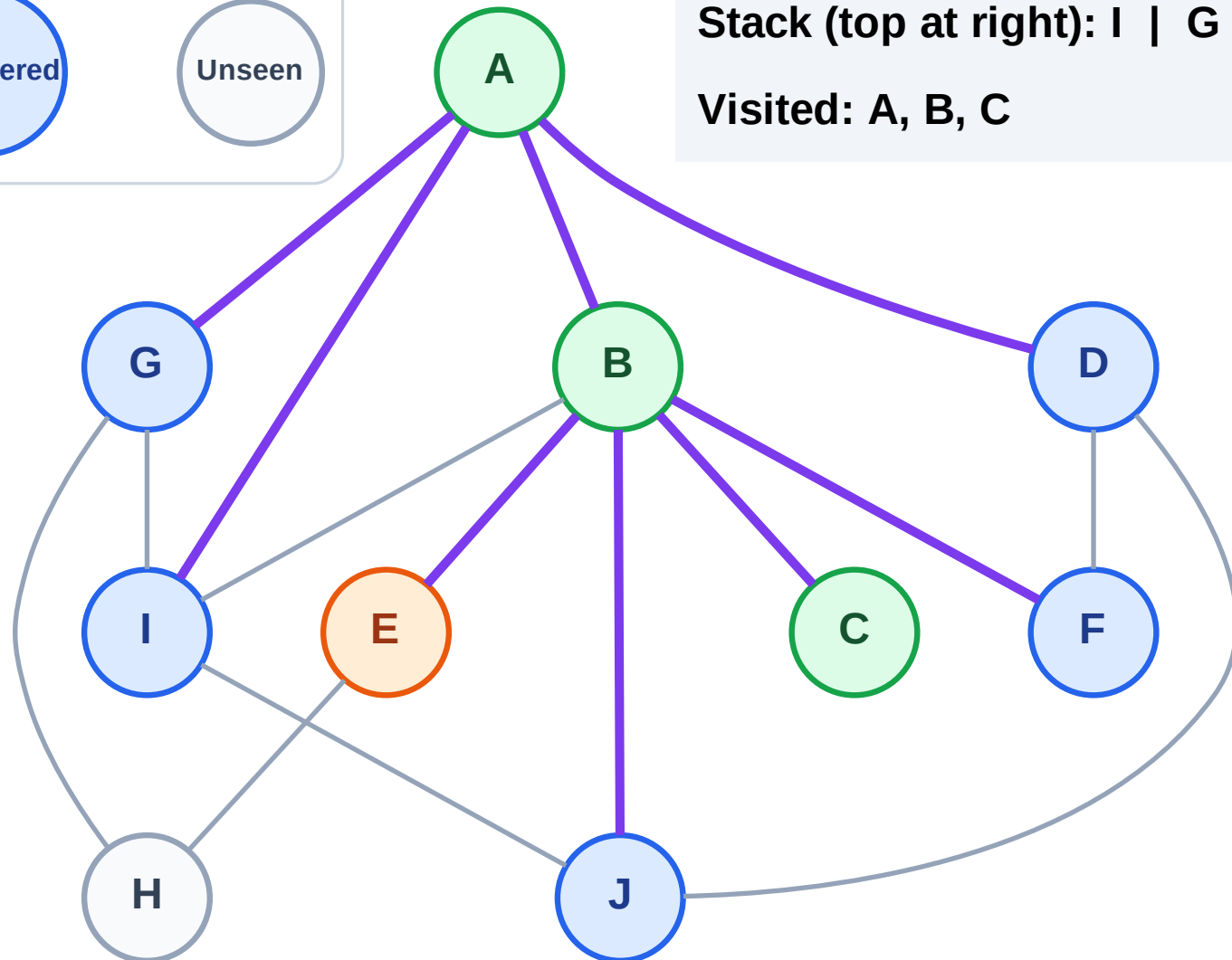
Discovered



Unseen

Stack (top at right): I | G | D | J | F

Visited: A, B, C



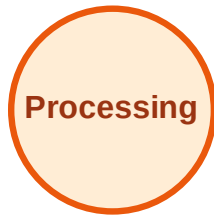
# DFS Traversal

Step 9: Discover H from E

## Legend



Complete



Processing



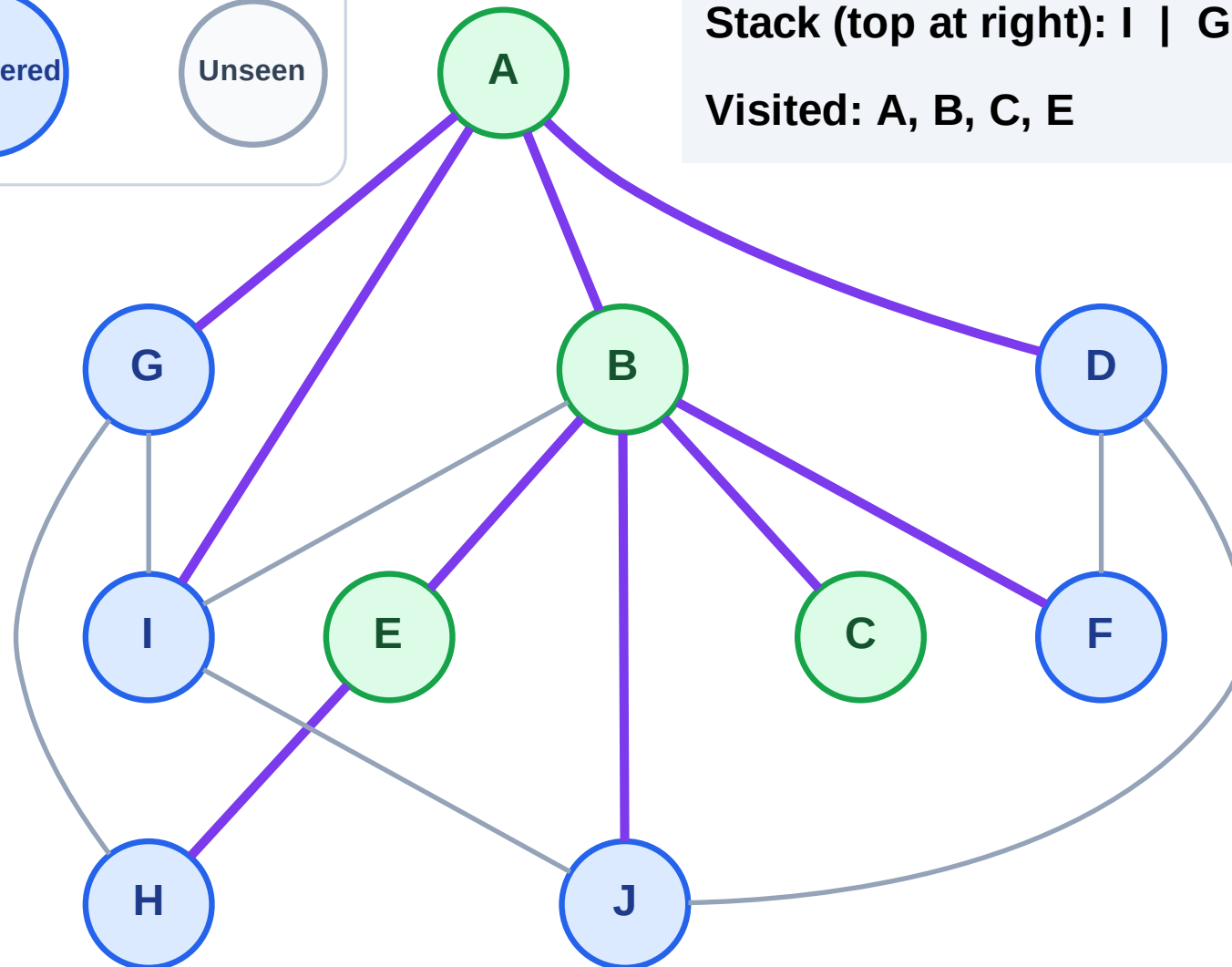
Discovered



Unseen

Stack (top at right): I | G | D | J | F | H

Visited: A, B, C, E



# DFS Traversal

Step 10: Process node H

## Legend

Complete

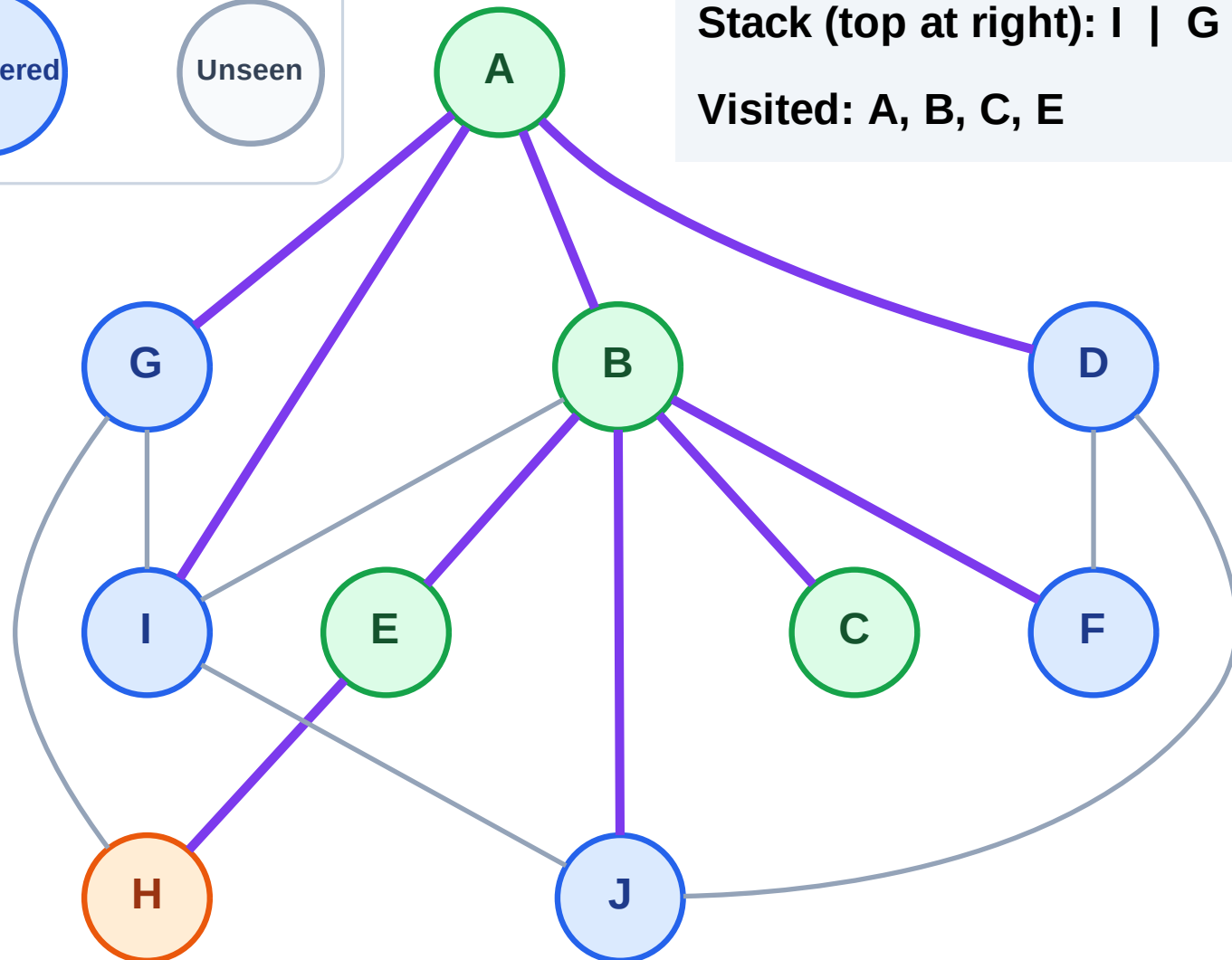
Processing

Discovered

Unseen

Stack (top at right): I | G | D | J | F

Visited: A, B, C, E



# DFS Traversal

Step 11: No new neighbors from H

## Legend

Complete

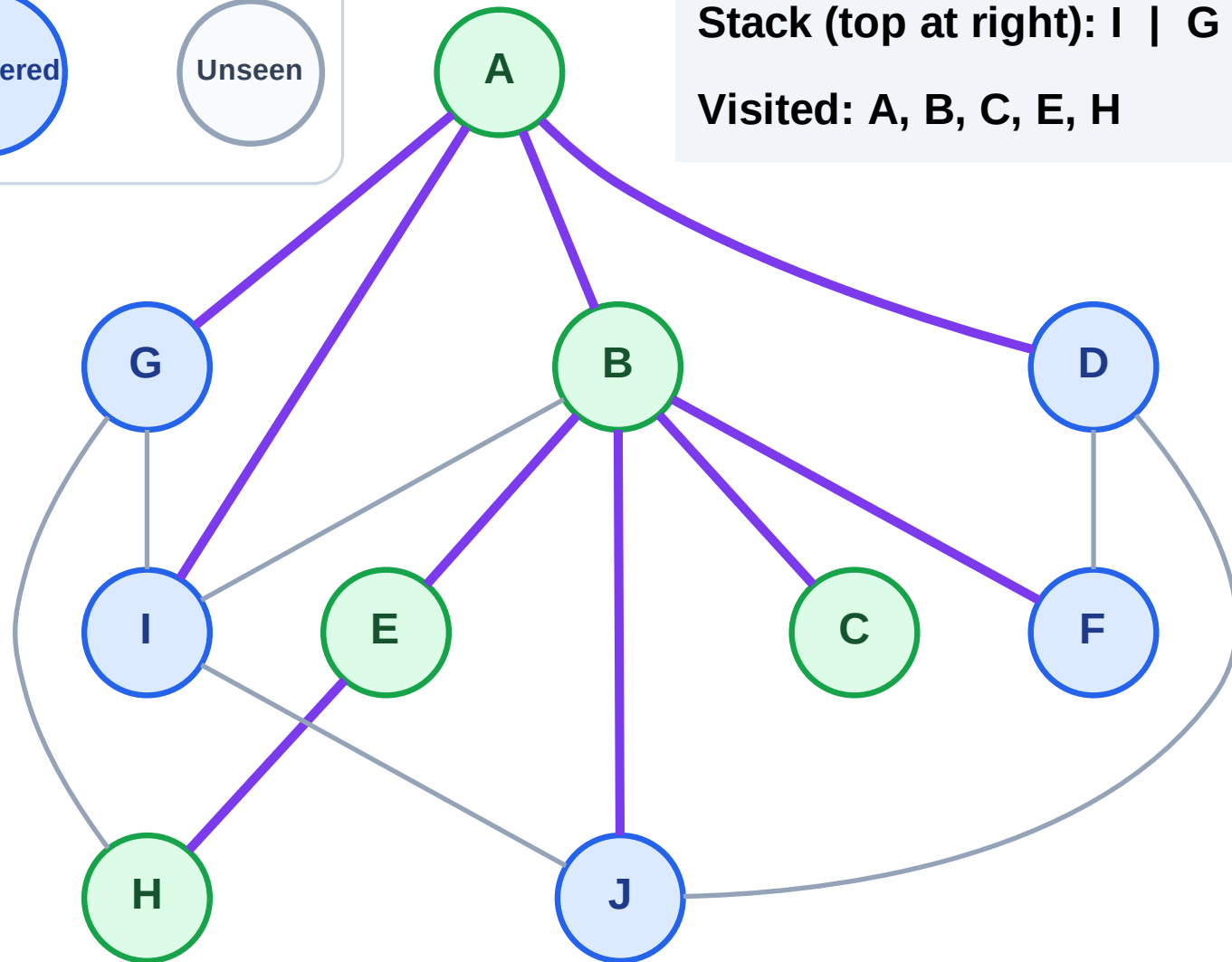
Processing

Discovered

Unseen

Stack (top at right): I | G | D | J | F

Visited: A, B, C, E, H

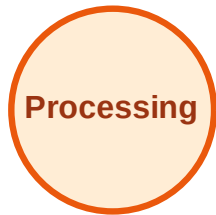




# DFS Traversal

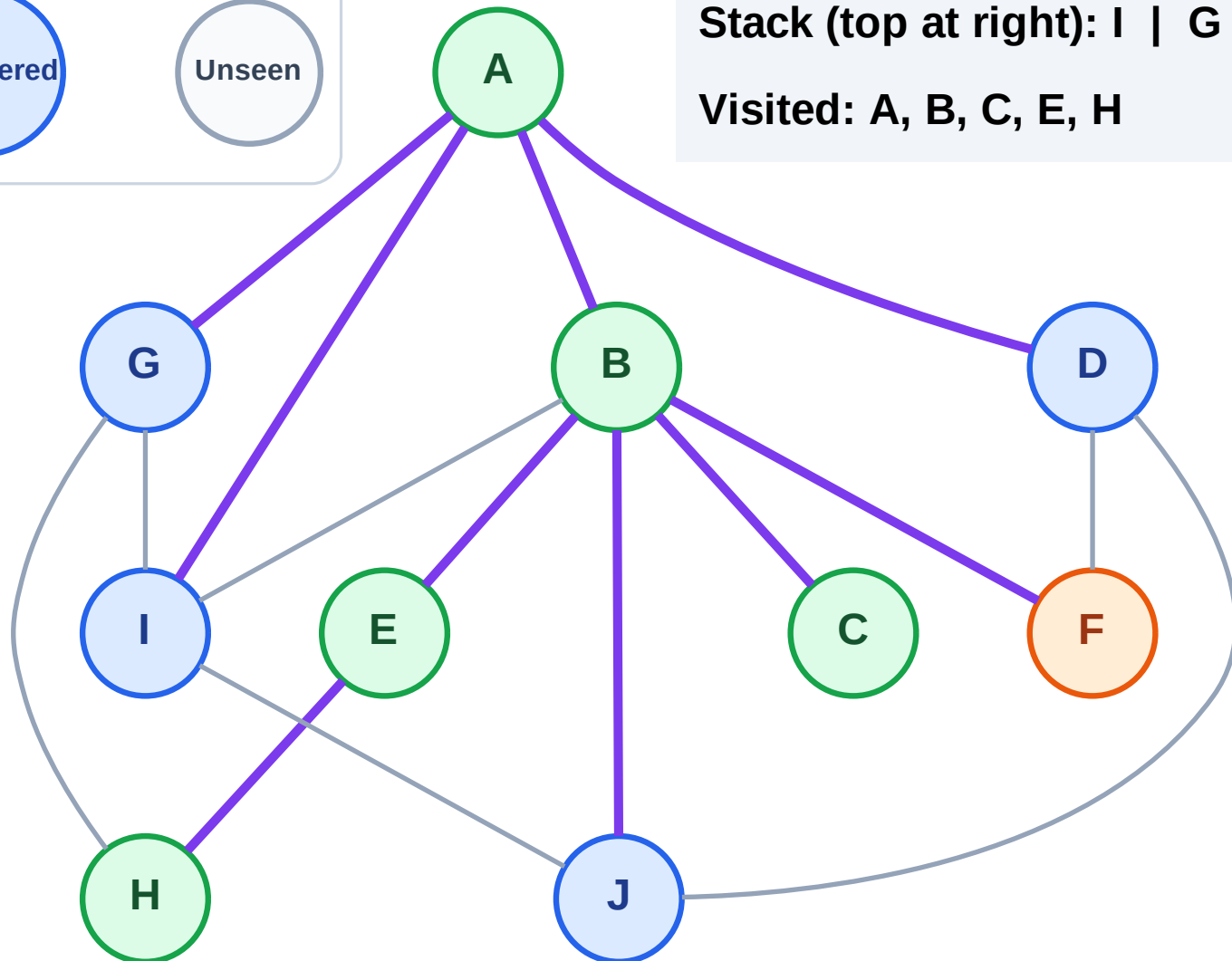
Step 12: Process node F

## Legend



Stack (top at right): I | G | D | J

Visited: A, B, C, E, H



# DFS Traversal

Step 13: No new neighbors from F

Legend

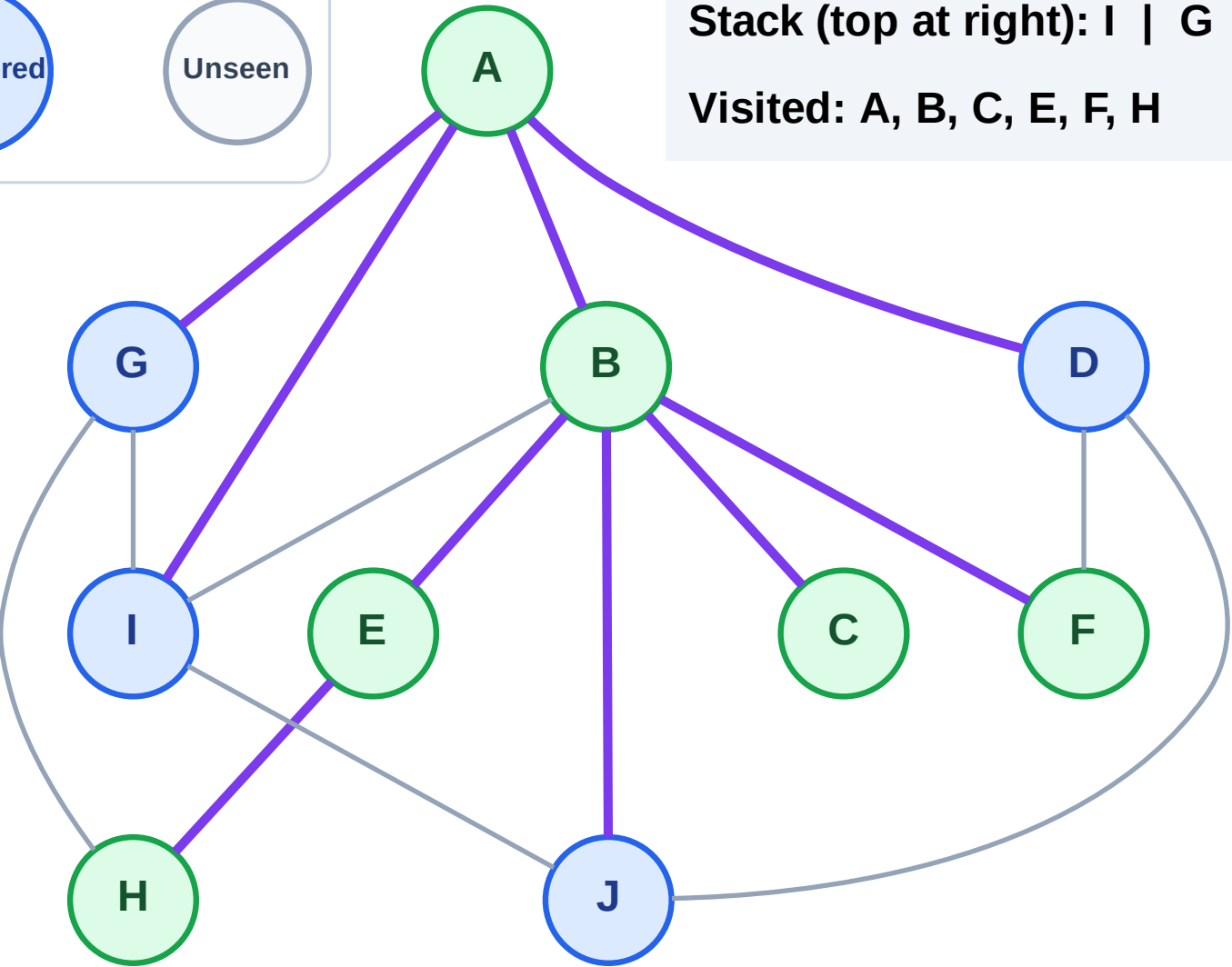
Complete

Processing

Discovered

Unseen

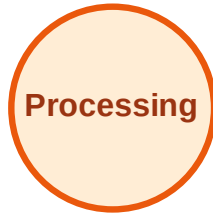
Stack (top at right): I | G | D | J  
Visited: A, B, C, E, F, H



# DFS Traversal

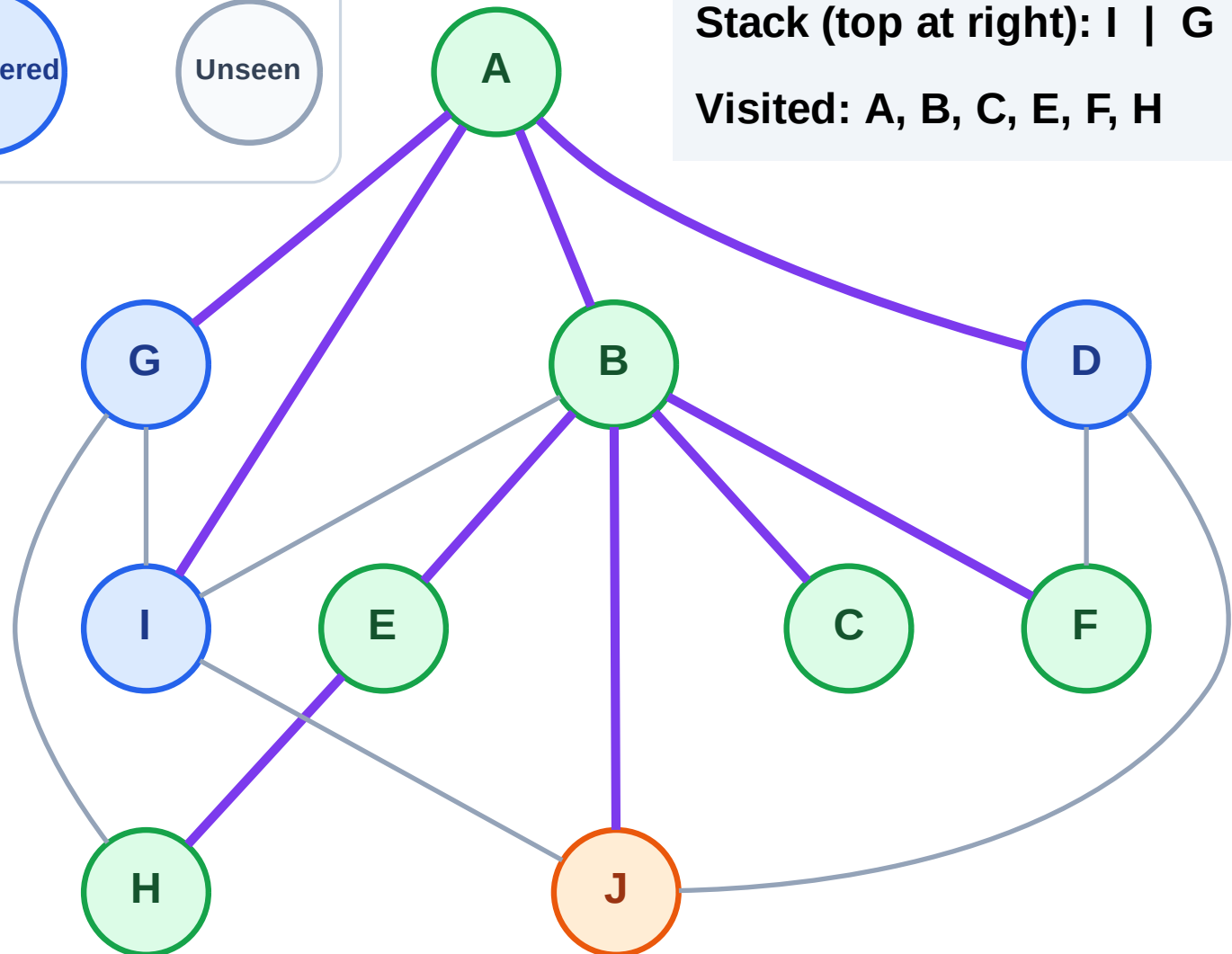
Step 14: Process node J

## Legend



Stack (top at right): I | G | D

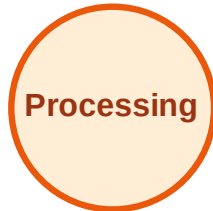
Visited: A, B, C, E, F, H



# DFS Traversal

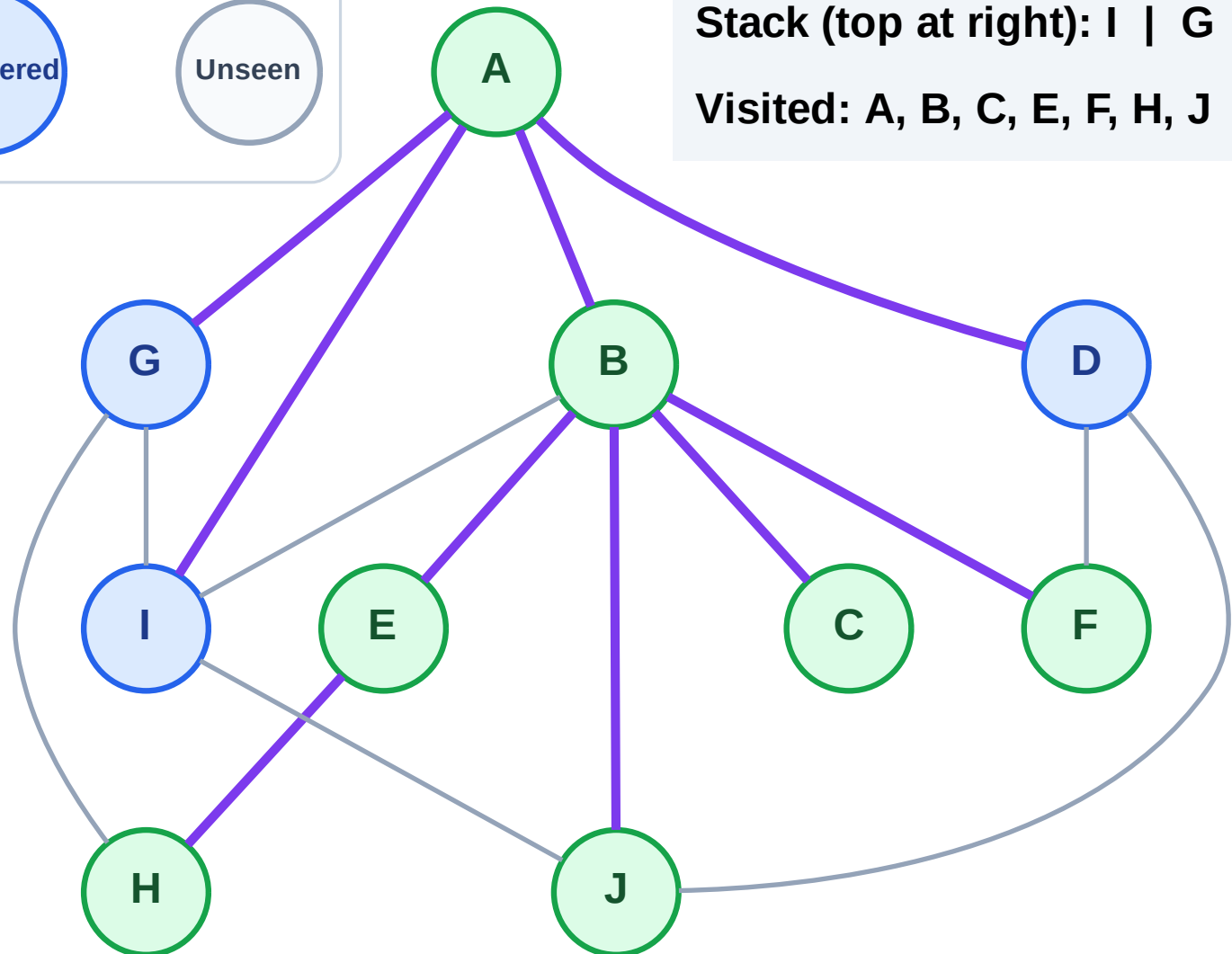
Step 15: No new neighbors from J

## Legend



Stack (top at right): I | G | D

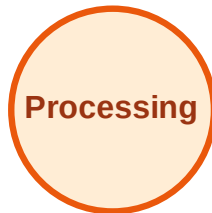
Visited: A, B, C, E, F, H, J



# DFS Traversal

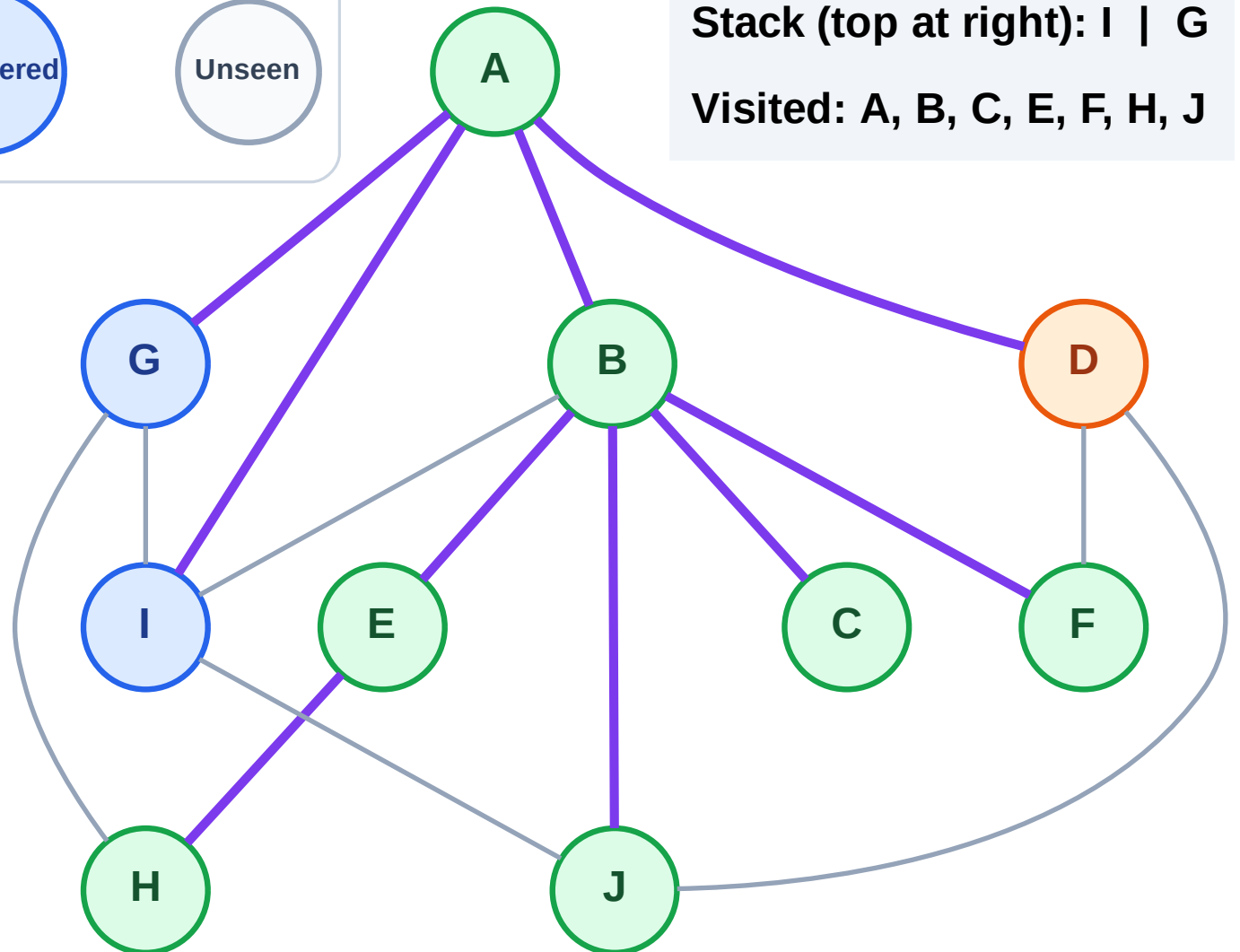
Step 16: Process node D

## Legend



Stack (top at right): I | G

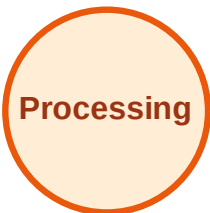
Visited: A, B, C, E, F, H, J



# DFS Traversal

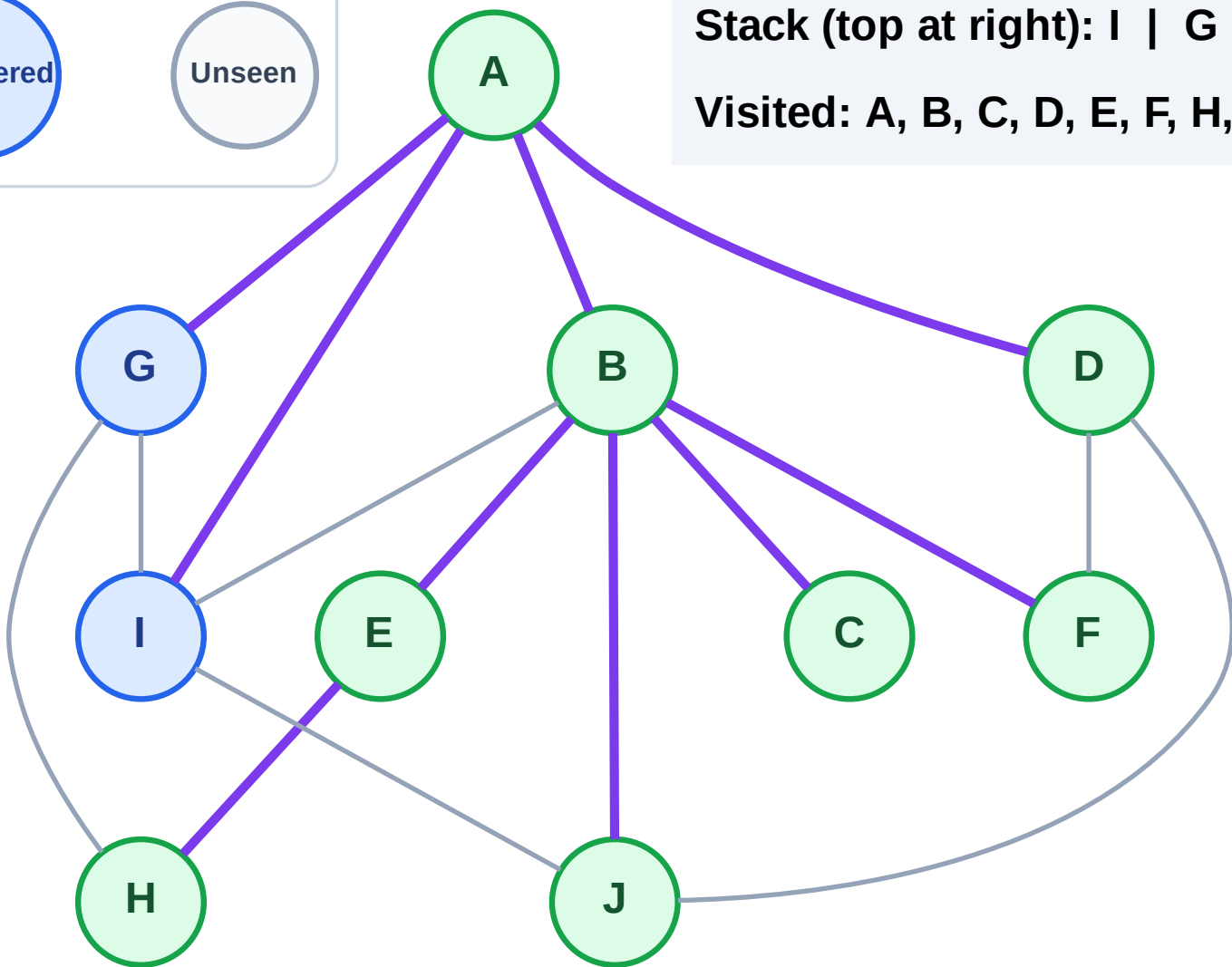
Step 17: No new neighbors from D

## Legend



Stack (top at right): I | G

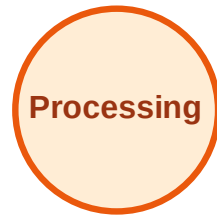
Visited: A, B, C, D, E, F, H, J



# DFS Traversal

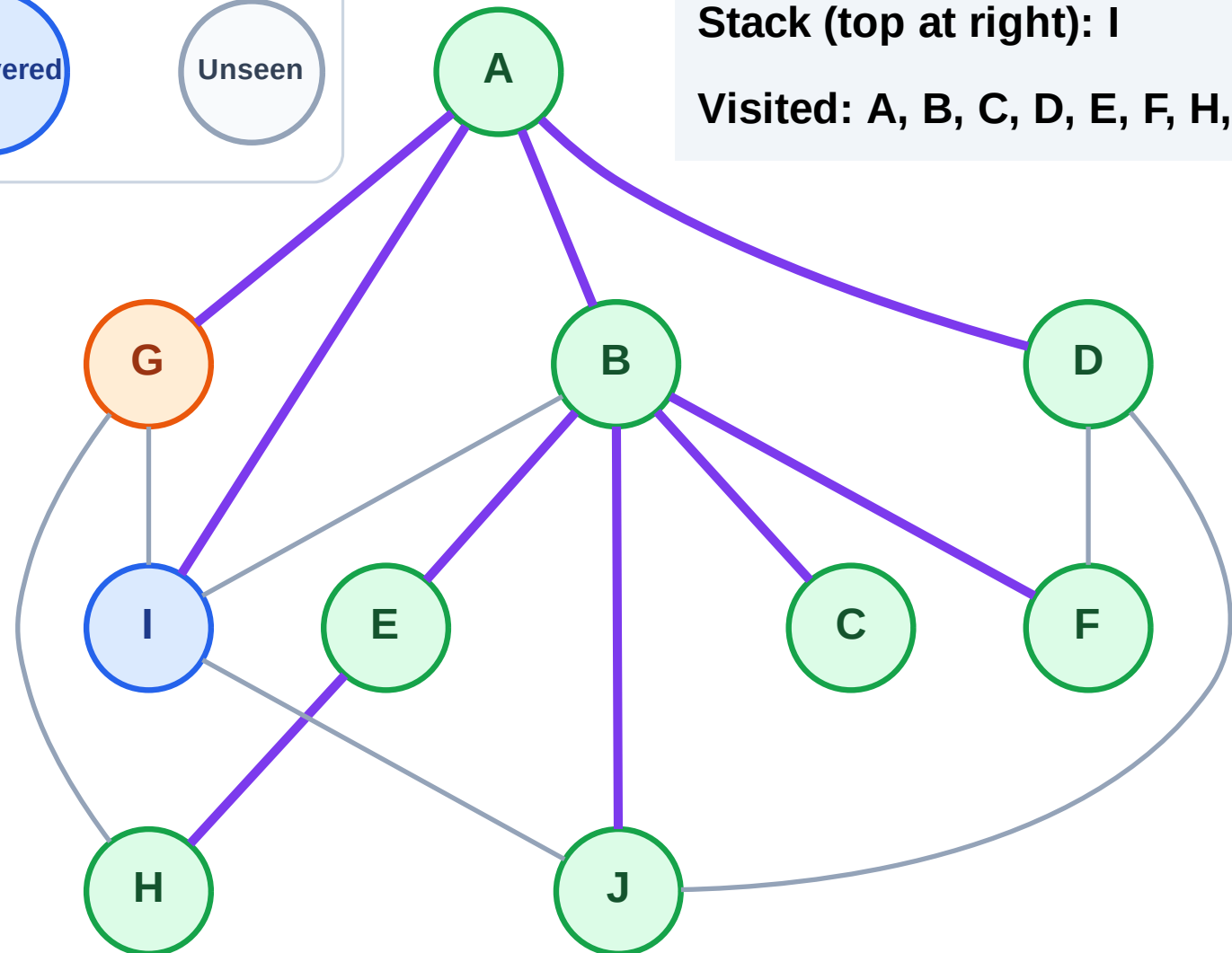
Step 18: Process node G

## Legend



Stack (top at right): I

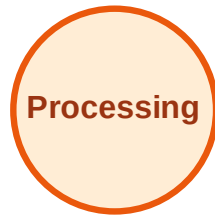
Visited: A, B, C, D, E, F, H, J



# DFS Traversal

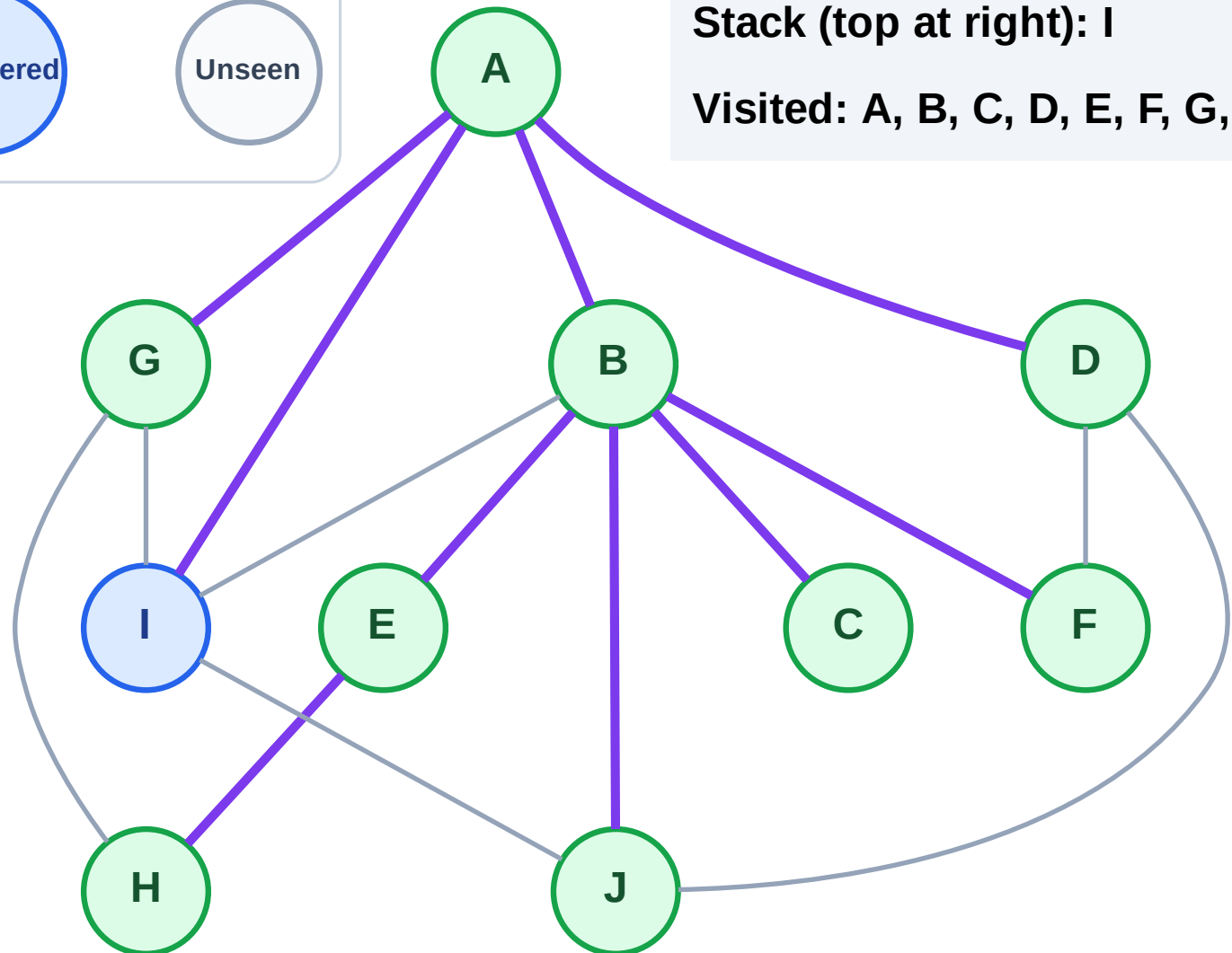
Step 19: No new neighbors from G

## Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J

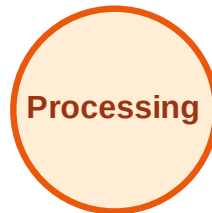




# DFS Traversal

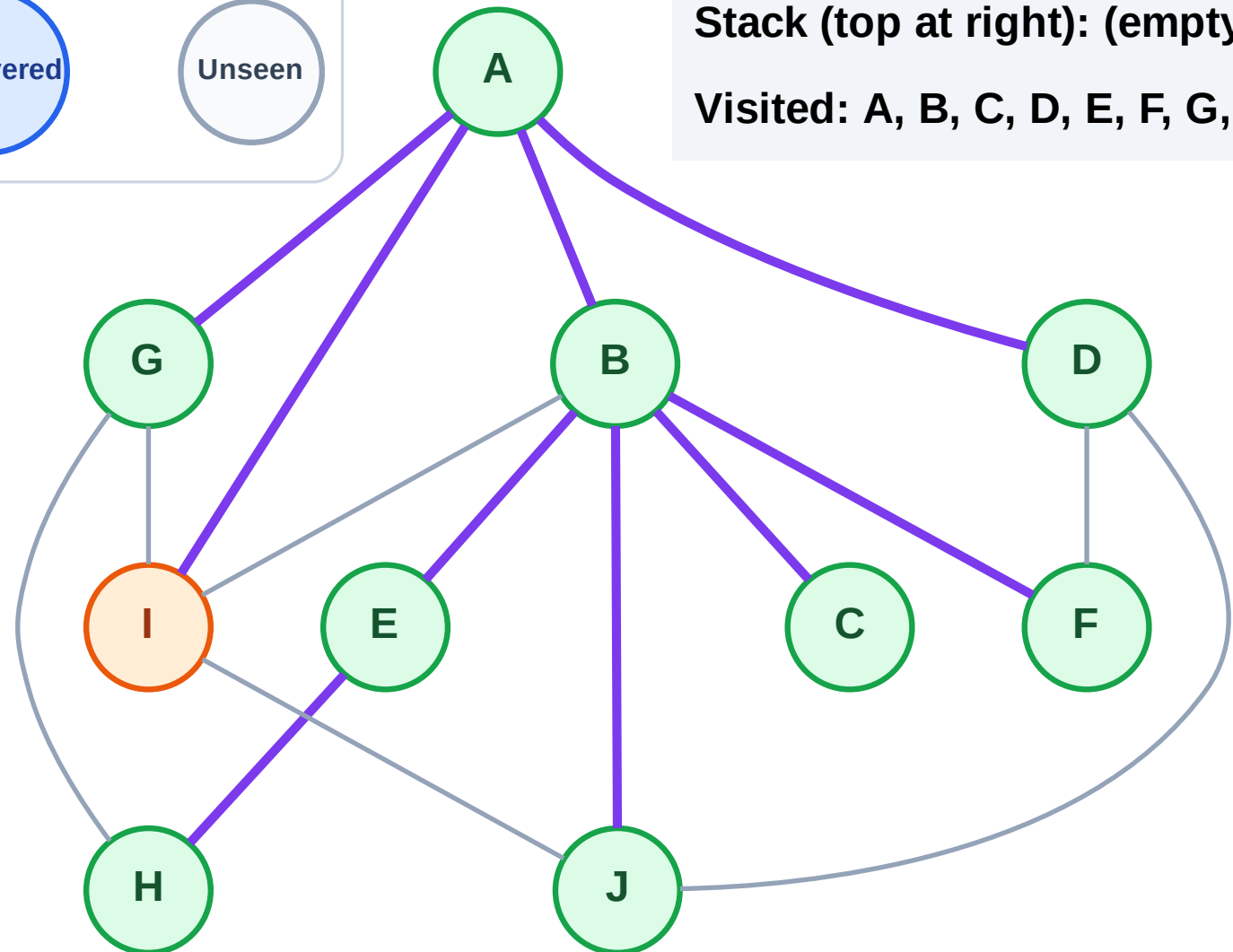
Step 20: Process node I

## Legend



Stack (top at right): (empty)

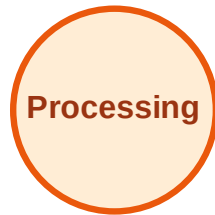
Visited: A, B, C, D, E, F, G, H, J



# DFS Traversal

Step 21: No new neighbors from I

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

